



An Optimized Federated Large Language Model-Based Multi-Agent System for Sybil Attack Detection in Software-Defined Vehicular Edge Networks

An undergraduate project proposal report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

**Degree of the Bachelor of the Science of Engineering
Honours**

by

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Abstract

Software-Defined Vehicular Edge Networks (SDVENs) are critical infrastructure for intelligent transportation, yet their open multi-tier architecture is acutely vulnerable to Sybil attacks in which adversaries forge multiple fake identities to inject false traffic data, manipulate routing, and corrupt distributed intelligence. Existing frameworks suffer five compounding limitations: FL-based detection achieves low accuracy using shallow models that lack contextual reasoning; all prior federated approaches assume honest RSUs and SDN controllers, leaving control-plane Sybil attacks undetected; no work formally quantifies mobility-induced threat amplification specific to SDVEN; the sole LLM applied to vehicular misbehavior operates in a centralized, non-federated setting; and no vehicular Sybil detection framework provides post-quantum cryptographic mitigation.. This paper presents DSMF-SDVEN, a Dual-Mode Signature-Characterized Federated Large Language Model Multi-Agent System with Post-Quantum Threshold-Dilithium and Kyber Mitigation for comprehensive Sybil attack detection and mitigation in SDVEN. The framework formally characterizes six Sybil attack variants—Outsider, Insider Direct Simultaneous, Insider Direct Non-Simultaneous, Insider Indirect, Malicious RSU, and Malicious SDN Controller—through novel attack signature formulations quantifying their observable behavioral deviations. Detection operates in two dynamically selected modes: a lightweight mode deploying rule-based signature threshold detection with no computational overhead, providing sub-10 ms response; and a full mode activating a privacy-preserving federated learning pipeline with ML models at the vehicle tier, three specialized LLM agents at the RSU edge tier, and MPC-secured global aggregation at the SDN controller tier. Mitigation employs CRYSTALS-Dilithium threshold ring signatures with t -of- n RSU co-authorization and CRYSTALS-Kyber key encapsulation for post-quantum session re-keying, with IPFS providing distributed tamper-evident evidence storage. The framework operates without trust assumptions on RSUs or the SDN controller, explicitly detecting Sybil attacks originating from compromised control-plane entities. Evaluation against three baselines using five metrics including Matthews Correlation Coefficient demonstrates.

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Acronyms

AI	Artificial Intelligence
ANN	Artificial Neural Networks
AoA	Angle of Arrival
CA	Certificate Authority
CH	Cluster Heads
CL	Centralized Learning
CNN	Convolutional Neural Networks
DL	Deep Learning
DoS	denial-of-service
DT	Decision Tree
ELM	Extreme Learning Machine
FL	Federated Learning
FLEMDS	Federated Learning Entrusted Misbehavior Detection System
GSIS	Group Signature and Identity-based Signature
IBV	Identity-based Batch Verification
ITS	Intelligent Transportation Systems
KNN	K-Nearest Neighbor
LLM	Large Language Model
LR	Logistic Regression
LSB	Least Significant Bit
LSTM	Long Short term Memory
ML	Machine Learning
NB	Naïve Bayes

NLP Natural Language Processing

OBU On-Board Unit

PKI Public Key Infrastructure

RNN Recurrent Neural Network

RSSI Received Signal Strength Indicator

RSU roadside unit

RSUs roadside units

SDN Software Defined Network

SDVEN Software Defined Vehicular Edge Network

SDVF software-defined vehicular fog networks

SDVN Software Defined Vehicular Network

SDVNs Software Defined Vehicular Networks

SHAP SHapley Additive exPlanations

SVM Support Vector Machines

TAN Transformer Based Attention

TCNN Transformer Convolutional Neural Network

VANET Vehicular Ad Hoc Network

VANETs Vehicular Ad Hoc Networks

VIN Vehicle Identification Number

Chapter 1

Introduction

1.1 Background

1.1.1 Vehicular Networks Evolution

Vehicular networks have evolved, meeting the increasing demands related to Intelligent Transportation Systems (ITS), autonomous vehicles, and real-time road safety applications. In the early days, vehicular communication was mainly based on Vehicular Ad Hoc Networks (VANETs). In VANETs, vehicles directly interact with other vehicles (V2V) and with the roadside Infrastructure (V2I) network forms automatically and changes dynamically. This enables all vehicles to broadcast important information, ranging from real-time traffic congestion to immediate hazard warnings. When we consider VANETs, since they are fully decentralized and vehicles move, they face several problems, such as frequent network changes, poor scalability, and difficulty in managing the network efficiently.

Software Defined Vehicular Network (SDVN) [1], were introduced to solve the above issues in VANETs by applying the concept of Software Defined Network (SDN) to VANETs were introduced to solve the above issues in VANETs. This SDVN divides the network into a data plane and a control plane. In here, a central SDN controller manages the network and makes decisions for routing and resource allocation. This centralized approach makes the network easier to manage. It is highly flexible and adapts to changing traffic conditions. It also enhances security and overall network performance compared to traditional VANETs.

As vehicular networks have to provide low delay and high data processing speed, edge computing was introduced to SDVN [2], leading to Software Defined Vehicular Edge Network (SDVEN). In SDVEN, computing and storage resources are placed closer to the vehicles using roadside units (RSUs) and edge servers. This reduces the need to send data to distant cloud servers, which helps reduce communication delays, and it helps in processing information faster. Thus, the SDVEN can be efficient. As a result, SDVEN can better support real-time applications that require rapid responses, including collision avoidance and autonomous driving.

In summary, the evolution from VANETs to SDVN and subsequently to SDVEN introduces various benefits, including centralized network control, network scalability, and reduced latency.

1.1.2 Security Challenges in SDVEN

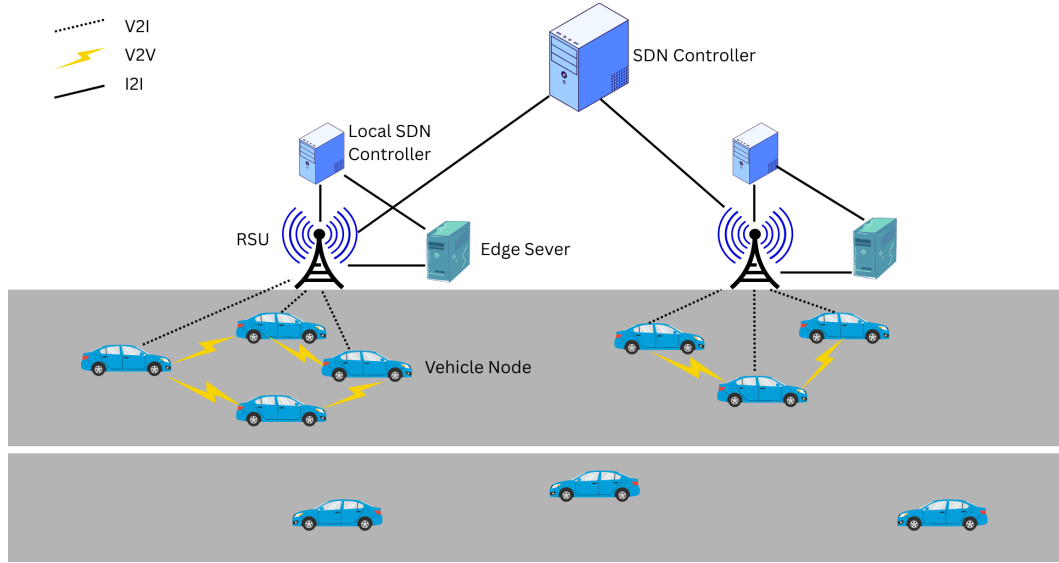


Figure 1.1: SDVEN Architecture Diagram

In traditional VANETs, the control plane and the data plane are tightly coupled within the same hardware. Since each vehicle independently made routing decisions under high mobility, the network struggled to adapt to rapid topology changes. To address this limitation, Software Defined Vehicular Networks (SDVNs) introduced a centralized controller. Vehicles mainly focus on data forwarding, while routing and control decisions are centralized at SDN controller. This architecture enables efficient management of rapidly changing vehicular connections. However, the physical separation between data-plane devices and the centralized controller can produce additional latency. Moreover, the centralized controller becomes a high-value target for cyber-attacks. SDVEN extend SDVN by integrating edge computing capabilities. By moving computation closer to vehicles, SDVEN reduces network latency and packet loss [3].

Despite these advantages, SDVEN environments still face several security challenges. Vehicles generate sensitive information, including location, speed and identity-related data. If this information is exposed, personal privacy may be violated, vehicles can be tracked overtime and behavioral patterns can be caught. Consequently, there exists trade-off between protecting driver privacy and maintaining accountability for safety and legal purposes [4].

Additionally, the open wireless communication medium used in SDVEN makes the network vulnerable to eavesdropping and message interception. SDVEN is also vulnerable to Sybil attacks, in which a malicious vehicle make multiple identities to disrupt network operations. Due to high vehicle mobility and short communication windows, establishing trust and verifying identities in real time remains a significant challenge [5].

Furthermore, edge nodes are in exposed environments and typically possess weaker security defenses than centralized cloud data centers, making them easy targets for attackers [6],[7].

1.1.3 Sybil Attacks

Sybil attacks represent one of the most challenging security threats in SDVEN. During a Sybil attack, a malicious entity creates and controls multiple fake identities, known as Sybil nodes, to exert a disproportionate impact on network operations [8]. The attack exploits the open nature of vehicular networks, which require vehicles to communicate with unknown peers. In a typical Sybil attack scenario, a malicious vehicle creates several fake identities that appear to be authentic network participants. By coordinating these fake identities, the attacker can overwhelm legitimate traffic, inject false data or influence network consensus.

Types of Sybil Attacks

Three of the most well-known Sybil attack variants are as follows [9].

1. Direct Vs Indirect Sybil Attacks

- (a) **Direct:** Every maliciously established Sybil node communicates and interacts directly with legitimate nodes.
- (b) **Indirect:** Sybil nodes communicate and interact with legitimate nodes only through the malicious node.

2. Insider Vs Outsider Sybil Attacks

- (a) **Insider:** A malicious node first gains legitimate access and then creates the Sybil nodes.
- (b) **Outsider:** A malicious node injects fake identities without being an accepted member of the network.

3. Simultaneous Vs Non-simultaneous Sybil attacks

- (a) **Simultaneous:** A malicious node establishes many Sybil nodes simultaneously at the same time.
- (b) **Non-simultaneous:** A malicious node uses multiple Sybil nodes sequentially, making them appear and disappear at different times.

This project will focus on all identified Sybil attack variants mentioned above, allowing a comprehensive and generalizable analysis.

1.2 Limitations of Existing Work

The decentralized, dynamic, and software-defined architecture of SDVENs introduces significant security vulnerabilities, with Sybil attacks emerging as one of the most severe threats. In Sybil attacks, malicious entities introduce multiple fake identities called Sybil nodes to inject false data. Using Sybil nodes, malicious entity could be able to distort network perception, inject false traffic information, manipulate network decisions and ultimately compromise road safety. [10] Conventional detection approaches including crystallographic certification, signal strength analysis, and centralized machine learning models, often fall short in resource-constrained real world vehicular networks including SDVENs. They struggle with real-time adaptability, scalability, latency and preservation of data privacy across distributed vehicular nodes [11]. The traditional approaches that offer limited contextual understanding of complex dynamic attack behaviours has been used in most of existing studies.

1.3 Problem Statement

Sybil attacks in SDVEN — amplified by vehicle mobility and executable across all tiers including compromised RSUs and SDN controllers — cause false traffic state injection and federated model poisoning that directly endanger road safety decisions, yet no existing framework provides mobility-aware, multi-tier Sybil detection with post-quantum-secured mitigation suitable for real-time SDVEN deployment.

1.4 Method Overview

Addressing the limitations of existing work, this project proposes an optimized Large Language Model (LLM)-based multi-agent system with Federated Learning (FL) architecture for real-time Sybil attack detection in SDVENs. By introducing FL to train LLMs collaboratively without raw data exchange, and integrating multi-agent coordination with optimization techniques for efficient resource allocation and model convergence, the system aims to deliver a privacy-preserving, adaptive defense mechanism suitable for practical deployment in evolving vehicular ecosystems.

1.5 Objectives and Scope

1.5.1 Objectives

1. To design and create a SDVEN simulation environment that enables intelligent and decentralized Sybil Attack detection.
2. To develop a privacy preserving FL framework that enables collaborative Sybil attack detection across vehicle nodes, roadside unit (RSU)s and the SDN controller.
3. To implement a multi-agent based detection mechanism that can recognize diverse Sybil attack behaviors in SDVEN environments.

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1.5.2 Research Questions

- RQ1.** Do mobility-aware attack signatures — incorporating velocity-dependent RSSI fading, trajectory continuity under topology churn, and handoff synchronization gaps — yield statistically superior detection accuracy over mobility-agnostic feature baselines across all six Sybil attack variants?
- RQ2.** Does contact-window-driven dual-mode selection achieve both sub-10 ms lightweight detection feasibility under high-speed vehicular conditions and full-mode detection accuracy comparable to always-on full-mode deployment, without degradation under varying mobility scenarios?
- RQ3.** Does the federated LLM multi-agent pipeline, operating under the privacy and resource constraints of SDVEN edge servers, achieve detection accuracy and variant classification precision superior to FL-only baselines without LLM reasoning?
- RQ4.** How does CRYSTALS-Dilithium t -of- n threshold revocation perform in terms of revocation latency and false revocation rate under adversarial conditions, including single and coordinated RSU compromise, relative to non-threshold mitigation baselines?

1.5.3 Project Scope

1. This project is limited to the simulation of a SDVEN using the NS-3 Simulator with vehicles, edge nodes, and SDN controller.
2. There is no physical implementation or hardware deployment.
3. Normal Traffic patterns and Various Sybil attack scenarios, such as Outsider Sybil Attacks, Insider Direct Simultaneous Sybil Attacks, Insider Direct Non-Simultaneous Sybil Attacks, Insider Indirect Sybil Attacks, Malicious RSU Sybil Attacks, and Malicious SDN Controller Sybil Attacks are generated for testing purposes.

4. The FL framework and LLM are also evaluated only in a simulated environment.
5. The number of edge nodes related to FL is limited because of the computational and simulation constraints.
6. Real-time Sybil attack detection performance is measured under simulated latency conditions.
7. Fundamental information security mechanisms such as digital signatures for validation, hashing to protect transferred data and encryption across plane communication serve as the foundation layer for reliability. Other security mechanisms are build upon this core mechanism.

1.6 Contributions and Novelty

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1. **Mobility-aware Sybil threat modeling:** Four mobility-induced SDVEN-specific threat amplification mechanisms — topology churn, RSU handoff synchronization gaps, short contact windows, and velocity-dependent RSSI fading — are formally quantified and integrated into the detection pipeline, establishing why SDVEN Sybil attacks are structurally more severe than in static VANETs.
2. **Formal mobility-cognizant attack signature characterization:** Six Sybil attack variants are formally characterized through novel signature formulations that incorporate mobility-dependent RSSI cross-correlation, DTW-based trajectory continuity under fading, propagation depth decay, and FL model divergence, providing the measurable detection basis for both operating modes.
3. **Dual-mode detection framework with contact-window-driven mode selection:** A dynamic mode-selection mechanism governed by available computational resources and RSU contact window duration T_c switches between lightweight rule-based signature detection (sub-10 ms, no overhead) and full ML-FL plus LLM multi-agent detection, directly exploiting the mobility model to ensure mode feasibility under high-speed vehicular conditions.
4. **Federated LLM multi-agent detection:** Three LoRA-adapted LLM agents at the RSU edge tier provide contextual Sybil attack variant reasoning, with LLM-FL propagating only quantized adapter updates to preserve edge-server feasibility.
5. **MPC-secured federated learning:** Shamir secret sharing enables MPC-based FL gradient aggregation among RSUs, preventing a compromised RSU from poisoning the global detection model without requiring a trusted aggregator.

6. **Post-quantum threshold revocation:** A CRYSTALS-Dilithium t -of- n threshold ring signature scheme co-authorized by multiple RSUs provides quantum-safe Sybil revocation robust to single-RSU compromise; CRYSTALS-Kyber KEM ensures post-revocation quantum-safe session re-keying.
7. **Control-plane threat modeling with IPFS evidence:** Sybil attacks originating from malicious RSUs and compromised SDN controllers are explicitly modeled and mitigated without any control-plane trust assumption; IPFS content-addressed storage signed with threshold Dilithium provides real-time tamper-evident evidence propagation without blockchain consensus overhead.

1.6.1 Novelty

This work is the first to ground SDVEN Sybil detection in a formal mobility threat model — quantifying topology churn, handoff synchronization gaps, contact windows, and fading-induced RSSI ambiguity — and to propagate this model directly into attack signature formulations, dual-mode contact-window-driven mode selection, and post-quantum threshold revocation urgency, all within a unified federated LLM multi-agent framework that eliminates control-plane trust assumptions, employs MPC-secure aggregation, and delivers IPFS-backed tamper-evident evidence without blockchain overhead.

1.7 Implications

The proposed optimized federated LLM-based multi-agent framework is expected to have several important implications for vehicular network security and related domains. This study aims to extend existing trust and attack detection in SDVENs by introducing a decentralized, cross-layer trust monitoring approach that explicitly considers malicious behavior in both data and control planes. This work is expected to contribute to the knowledge on distributed intelligence and privacy-preserving learning in highly dynamic network environments by integrating federated learning with multi-agent LLMs.

The proposed framework is expected to provide a detecting mechanism for many Sybil attack scenarios, including outsider, insider simultaneous, insider non-simultaneous, and insider indirect attacks, as well as malicious RSU and SDN controller behaviors. By combining authentication mechanisms, behavioral monitoring, Received Signal Strength Indicator (RSSI)-based analysis, and data similarity evaluation, the proposed system is anticipated to enhance real-world usage of secured vehicular networks while preserving scalability and data privacy. Therefore, this framework could be applied in intelligent transportation systems to improve trustworthiness, safety, and resilience against coordinated attacks.

The outcomes of this research are expected to influence future security guidelines and architectural decisions for software-defined vehicular edge networks. Particularly, centralized trust assumptions are no longer valid. The decentralized trust monitoring across network entities may support the development of more resilient standards for next-generation vehicular infrastructures. This aspect may open new directions for applying distributed trust monitoring to fraud-prone domains.

Chapter 2

Literature Review

2.1 Previous Work

2.1.1 Traditional Approaches

1. Resource Testing Method

Resource testing is a basic method to detect Sybil attacks by determining whether each device has typical physical limitations. This concept was initially proposed for peer-to-peer networks, proposing to test computational power, storage capacity, and communication capabilities [12]. The first approach of this paper used computational puzzles. It was a local device that created a random value and asked each identity to solve a math problem within a time limit. The second method involves storage challenges, where each identity must save a large amount of unique data that can not be compressed. Here, the checker randomly asked to see part of this data to make sure it is really stored. The third approach used a communication timing test. It means sending out requests and measuring how long responses take to reply to those requests when those replies came from different physical locations based on the signal propagation. This traditional resource testing does not work well in sensor networks. Because attackers often use powerful devices, Testing computation power and storage fails. One of the problems with broadcasting requests and timing responses as a method of testing communications is network congestion many nodes reply all at the same time.

So new **radio resource testing method** for sensor networks was introduced in [9] to address these limitations. In this approach, they use a simple principle: A radio can not send or receive on multiple channels simultaneously. The test works by assigning different radio channels to neighboring nodes and listening to one channel randomly. If a node does not transmit on its assigned channel, it's likely a Sybil node. Performing this test several times increases the chance of detection it more possible.

2. Position Verification

Sybil attacks are a critical problem in computer networks, because attackers create many fake identities to harm the system. Position verification is the method that can identify fake identities by checking if nodes are really

where they claim to be. According to the traditional methods mainly used radio signals as mentioned above to verify the positions. The RSSI method, studied in [13], measures how strong radio signals are to estimate distances between devices. But in here, this method has a problem because walls and other devices can block the signals this system collects signal strength measurements from several special nodes that know their exact positions. These nodes are called anchor nodes. By combining measurements of multiple anchors, the system can find where a target node is located through a process called triangulation. Then it compares the node's claimed position with the measured position if they match.

A new approach that addresses a critical vulnerability in the position verification system was introduced in the paper [14]. This approach is a secure localization method using steganography combined with the Angle of Arrival (AoA) positioning technique. Steganography is the art of hiding information. this system operates in a $100\text{m} \times 100\text{m}$ area with Static Cluster Heads (CH) and mobile sensor nodes. They have known positions. Nodes obtain positions via AoA. CHs embed positions into cover images using Least Significant Bit (LSB) steganography. After that nodes extract data by comparing the stego-images with the originals. Only nodes possessing correct cover images can decode positions. This method secures military operations, emergency rescue missions.

3. Cryptography Method

Cryptographic methods are also widely used as a defense against Sybil attacks in vehicular ad hoc networks because they provide strong authentication and identity verification. Public Key Infrastructure (PKI) is a commonly used approach in this technique, where a trusted Certificate Authority (CA) assigns each vehicle a unique digital certificate. In this study, demonstrated that PKI-based authentication can prevent identity forgery by associating cryptographic credentials with physical vehicles[15]. However, PKI has some limitations because PKI schemes are vulnerable to insider attacks, as a compromised legitimate vehicle.

To address the limitations of the traditional PKI-based approaches, Group Signature and Identity-based Signature (GSIS) [16] was proposed. It is a new security protocol design for VANETs. Their protocol uses group signatures for communication between vehicles and identity-based signatures for communication between roadside units and vehicles. This combination solved the difficult problem of keeping vehicle identities for privacy, but allowed authorities can trace them. In this GSIS protocol, the driver's privacy is protected during normal driving because this means any receiver can verify that the message comes from a legitimate group member without knowing the exact vehicle that sent the message. This GSIS is much more efficient than the anonymous certificate approach previously.

For improving verification speed in high-density traffic scenarios, a scheme was introduced that allows RSUs to verify signatures simultaneously [17]. It is Identity-based Batch Verification (IBV). This scheme uses tamper-proof devices with identity-based cryptography installed in vehicles. These

temper-proof devices automatically generate random pseudo-identities and a matching private key for each message. It protects driver privacy and ensures efficient operations. Here, not checking each signature separately, like in traditional schemes, IBV can verify n signatures using only 3 pairing operations, no matter how many signatures are in the batch. In addition, IBV reduces communication, lowering bandwidth consumption.

2.1.2 Machine Learning Approaches

Machine Learning (ML)-based techniques have gained huge attention for Sybil attack detection in vehicular networks because of their ability to learn complex and dynamic behavioral patterns from vehicular data. Traditional rule-based security mechanisms are often ineffective in identifying Sybil nodes in highly dynamic vehicular environments, where attackers can continuously change identities and behavior. As a solution, supervised machine learning models including Support Vector Machines (SVM), Decision Trees, Random Forests, Artificial Neural Networks (ANN) and ensemble methods have been explored and tested to classify Sybil nodes in a vehicular network.

1. Behavior based ML approaches

Behavior based ML approaches are one of most common ML related strategies used for Sybil attack detection in vehicular networks. These methods rely on extracting and analyzing features such as vehicle position, speed, acceleration, trajectory consistency and message timing to identify abnormal patterns. In [18], authors proposed a behavior based distributed machine learning approach for Sybil attack detection in VANETs using an AdaBoost Classifier. And also, they extracted and analyzed vehicular motion patterns, including position, speed, acceleration and predicted location using VeReMi dataset, demonstrating that ensemble learning techniques can effectively distinguish Sybil nodes from legitimate vehicles by capturing inconsistencies in the driving behavior. A notable strength of this approach is the elimination of channel congestion issues inherent to centralized architectures as each vehicle performs local detection. However, the preprocessing phase requires matrix construction and eigenvalue computation and due to that computation is time-intensive. This limitation could be addressed through computational offloading to edge nodes or controllers in SDVEN implementations.

Furthermore, in [19], authors introduced a ML based Sybil attack detection technique using an Extreme Learning Machine (ELM). This method analyzes vehicle mobility features from periodic beacon messages including vehicle position, speed, timestamp, acceleration and acceleration difference and constructs a motion matrix for each vehicle. An ELM classifier is then employed to identify Sybil nodes by detecting similarities in mobility behavior multiple identities. This approach has improved the detection accuracy with low computational overhead. But the model relies on centralized data collection and does not explicitly account for control and data plane separation of SDVEN.

2. Collaborative ML approach

In addition to single model behavior based ML approaches, in [20], authors proposed a collaborative machine learning technique based on Majority Voting for Sybil attack detection in VANETs. This research employed five supervised ML classifiers, including K-Nearest Neighbor (KNN), Naïve Bayes (NB), Decision Tree (DT), SVM and Logistic Regression (LR) working simultaneously as base classifiers. In this study, both hard and soft majority voting mechanisms were evaluated and final outcome showed that majority voting soft can better detect the Sybil attack with high accuracy as compared to majority voting hard. However, the computational overhead of maintaining multiple classifiers present limitations. This ensemble framework could be efficiently hosted in the centralized controller in SDVEN implementation, utilizing the separation of control and data planes to divide the computational load while preserving real-time detection capabilities.

2.1.3 Deep Learning Approaches

Classical ML approaches have achieved significant success in Sybil attack detection in vehicular networks through various effective feature engineering and ensemble methods. But they often require manual feature extraction and may struggle with complex, non-linear patterns in high dimensional vehicular data. Deep learning approaches have evolved as powerful alternatives that can automatically learn hierarchical feature representations from raw data, potentially detecting more subtle behavior anomalies that typical machine learning methods might fail to recognize. In the context of SDVN and SDVEN architectures, deep learning models emerged to address these limitations by utilizing the computational resources of logically centralized controller to process large-scale traffic data and extract complex patterns to detect Sybil attacks.

1. Hybrid deep learning models architecture

Representing this paradigm shift toward deep learning based detection in SDVN architectures, in [11], authors proposed a comprehensive AI-driven two phase Sybil attack detection framework that strategically integrates Convolutional Neural Networks (CNN) and Long Short term Memory (LSTM) architectures to overcome the limitations of traditional ML approaches. During the phase I, a lightweight trust-based preliminary screening mechanisms are applied to filter the legitimate nodes and during phase II, hybrid 2-CNN 2-LSTM deep learning models deployed at the centralized SDVN controller are used to classify flagged suspicious vehicles into six categories. The CNN layers extract spatial features from 2D vehicle position grids detecting topology anomalies, while LSTM layers model temporal behavior sequences identifying trajectory inconsistencies. This study demonstrates superior detection accuracy through automatic feature learning, eliminating the manual feature engineering bottleneck that constrained previous methods. However, despite high detection accuracy, critical limitations such as elevated detection latency in real-time requirements limit practical deployment.

Similarly, in [21], authors proposed FOSA-DETECT, a deep learning framework for Sybil detection in VANETs, aimed at enhancing traffic efficiency through accurate identification of malicious identities. In this approach,

while Jelly Fish Optimization was used to efficiently capture the treats and a hybrid architecture of CNN and LSTM was used to classify normal/attacked classes. This hybrid optimization-DL approach advances prior works by emphasizing lightweight feature engineering for real-time ITS applications, though its reliance on VeReMi’s simulated urban traces limits validation in diverse real-world scenarios, and JFO’s stochastic nature may introduce variability in high-density networks.

Furthermore, in [22], authors extended deep learning based Sybil detection by proposing a hybrid IDS that clusters vehicles using Glowworm Swarm Optimization based K-Harmonic Means (GSOKHM) for mobility aware grouping and select cluster heads via the Floyd-Warshall algorithm to optimize shortest paths. A Chaotic Maps Elephant Herding Optimization enhanced Convolutional Neural Network (CMEHA-CNN) was used to detect the malicious cluster heads by refining features via PCA and tunes weights for binary classification. Then SHA2-Elliptic Curve Cryptography (SHA2-ECC) was used to securely transmit the legitimate cluster head data to the cloud and it enhanced the resistance to collisions and interception. Although this approach achieves high accuracy and outperforms baselines, it exclusively relies on highway scenarios, overlooking real world urban complexities and it exhibits high computational overhead due to its iterative chaotic maps and elephant herding optimization processes.

2.1.4 Large Language Models in Vehicular Network Security

The applications of LLMs in vehicle network security represents a growing research area that combine AI and Natural Language Processing (NLP) technologies for network security mechanisms in vehicular networks. LLM bases researches have been popular rapidly with the fast growth and the high usage of the acLLM based ai tools. The potential of acLLMs for detecting various types of network intrusions can be seen in recent researchers in the field of network security.

1. Transition from Recurrent Neural Network (RNN)s to Transformer Architectures

Early researches in vehicular network security were primarily focus in RNN and LSTMs approaches to analyze node behaviour using temporal features, but these models are following step by step analysing method that feature extraction relies only on previous hidden states, It creates processing bottleneck on those systems. Authors of [23] have addressed these limitations by proposing Transformer Based Attention (TAN) that uses self-attention mechanism other than recurrent layers. This architecture had achieved near perfect results with detecting fuzzy attacks and spoofing attacks. Main specialties shown in this research are by exploiting attention scores the system could identify the malicious messages without requiring a larger dataset of labeled messages, through they developed the solution for a specific vehicle model, with transfer learning this attention based model can be also used for other vehicle models.

2. Advancements in Vehicular Ad Hoc Network (VANET) Security and Explainable Transformer Models

Considering the high mobility and dynamic nature of autonomous vehicles VANETs has become a popular topology for vehicular network. But with this dynamic topology vulnerability to identity-based threats like Sybil attacks have increased. The authors of [24] has addressed these challengers with developing a Transformer Convolutional Neural Network (TCNN) to identify various malicious behaviours using sequential traffic data. The proposed system has identified multiple Sybil attacks including Data replay , DoS random and Grid Sybil. The TCNN has significantly enhanced the detection of complex fake identity attacks other than a traditional CNN based solution. Other speciality in the solution is SHapley Additive exPlanations (SHAP) technology is integrated with CNN to improve the explainability that makes model’s decisions interpretable for practical deployment ready in VANET by addressing the black-box nature of the deep learning approaches.

The first application of a large language model specifically for vehicular misbehavior detection is presented in [?], where Mistral-7B is fine-tuned using LoRA adapters on Basic Safety Message (BSM) sequences from the VeReMi dataset, achieving 98% binary and 96% multi-class detection accuracy while updating only 0.012% of model parameters. While demonstrating LLM viability for vehicular security, this work operates in a centralized, non-federated setting, employs no multi-agent coordination, and does not target SDVEN-specific attack variants such as those originating from compromised RSUs or SDN controllers.

Advanced Authentication Protocols for Vehicular Networks

Authentication protocol design directly informs the identity management and mitigation components of this work. In [?], an authentication protocol using zk-SNARKs and ECDH is proposed for IoV, achieving 20% lower computation overhead and 75% faster re-authentication. However, ECC-based primitives are vulnerable to quantum adversaries. In [?], a lattice-based Ring-LWE authentication protocol with ephemeral pseudonymization and spatial cloaking achieves 44.96% lower computation cost and formal AVISPA-verified resistance to Sybil, replay, and de-anonymization attacks. These works motivate post-quantum cryptographic integration in our mitigation layer.

2.1.5 Federated Learning in Vehicular Networks

FL is an emerging paradigm that enables multiple devices to collaboratively train machine learning models without sharing raw data, thereby preserving privacy and reducing the communication overhead associated with Centralized Learning (CL). In vehicular networks, where vehicles generate large amounts of sensitive and high velocity data, FL is particularly suitable for several applications such as autonomous driving, traffic prediction, vehicular object detection and road safety prediction. Unlike CL, which requires continuous transmission of raw data to a central server, FL allows vehicles to process data locally and share only model updates, thereby reducing latency and bandwidth requirements while improving real-time decision-making [25].

Recent studies have demonstrated the feasibility of FL in vehicular networks. For example, FL has been applied to image-based object detection, traffic flow prediction and autonomous driving, showing good performance in terms of accuracy and communication efficiency [25],[26]. These studies also highlight key challenges in deploying FL for vehicular networks, including heterogeneous computational resources, variable network connectivity, model training efficiency and secure aggregation of model updates [25],[26].

FL has recently gained attention as a good solution for security and attack detection in VANETs, owing to its ability to enable distributed model training while preserving data privacy and reducing communication overhead. FL-based threat detection frameworks deployed at the edge of VANETs have demonstrated effectiveness against various attacks, including false data injection, Sybil attacks, and denial-of-service (DoS) attacks, while maintaining real-time performance and privacy preservation [27].

Several studies have specifically investigated FL-based Sybil attack detection in vehicular environments. Trust-oriented federated frameworks have been proposed to detect Sybil attacks through decentralized trust validation and behavioral consistency analysis and achieving detection precision and accuracy of 98.5% and 98.8% respectively in VANET scenarios [28]. These approaches highlight the potential of FL to enhance security without relying on centralized data collection or heavy cryptographic mechanisms.

More advanced architectures have incorporated hierarchical federated learning to further reduce detection latency and improve scalability. For instance, FL-based Sybil detection systems employing three-tier model aggregation across RSUs, roadside controllers, and SDN controllers have achieved detection accuracies up to 98.7% with ultra-low latency of approximately 20 ms, demonstrating the feasibility of real-time security enforcement in SDVN [29]. Similarly, the Federated Learning Entrusted Misbehavior Detection System (FLEMDS) utilizes three-level model aggregation and on-vehicle Artificial Intelligence (AI) to perform local Sybil attack detection in software-defined vehicular fog networks (SDVF), achieving an accuracy of 87% in 6G-enabled environments [30].

Despite the demonstrated effectiveness of FL for security and Sybil attack detection in vehicular networks, existing approaches exhibit several critical limitations that restrict their real-world applicability and robustness. Most current FL-based frameworks rely on conventional ML or lightweight Deep Learning (DL) models that are task-specific and lack of capacity to capture complex, temporal, and semantic relationships contain in highly dynamic vehicular environments. These models are generally unable to catch advanced reasoning and natural language understanding capabilities that could enhance attack pattern recognition and adaptability. A critical oversight in these frameworks is their main assumption of aggregator integrity, with RSU controllers and SDN controllers presumed trustworthy despite the realistic threat of Byzantine or compromised entities that could poison global model updates and reduce detection accuracy.

Additionally, the absence of multi-agent coordination frameworks restricts scalability and collaborative decision-making of distributed security systems. These limitations constrain the robustness, adaptability, and deployment readiness of existing FL-based vehicular security frameworks and motivating the need for more advanced and trust-aware architectures.

Signal-Characteristic and Position-Based Detection

Physical-layer features are widely used as Sybil detection signals. RSSI-based detection methods exploit the principle that co-located Sybil nodes share correlated signal characteristics. In [?], RSSI time series are treated as a vehicular *voiceprint* and similarity among received series is computed for lightweight distributed detection. However, this approach is vulnerable to power-controlled attackers who deliberately vary transmission power to spoof distinct RSSI signatures. In [?], five Sybil attack models under power control are characterized and a linear SVM classifier is applied to distinguish attack patterns, demonstrating that naive RSSI methods fail against adaptive adversaries. More recently, [?] exploits the spatial structure of channel frequency response (CFR) profiles using signal clustering to identify co-located Sybil identities, validated through real DSRC V2V and V2I experiments. In [?], a multi-source data fusion detection (MDFD) framework characterizes Sybil behavior across four dimensions — original motivation, behavioral characteristics, resulting traffic-flow impact, and comparative validation — achieving 97.69% average detection accuracy for compound attacks. These physical and data-fusion approaches, while effective in specific scenarios, operate in flat VANET architectures without SDN-layer coordination or federated learning.

Blockchain and Cryptographic Sybil Mitigation

Blockchain provides tamper-resistant identity records that complement detection mechanisms. In [?], Sybil attacks are detected using proofs of work and location; vehicles solve computational puzzles unique to each RSU, preventing a single physical node from solving multiple puzzles simultaneously, achieving high detection rates with acceptable overhead in IEEE Transactions on Dependable and Secure Computing. In [?], a proof-of-location (PoL) scheme uses RSU-maintained smart contracts as ground truth for location verification, mitigating Sybil injection while preserving vehicle privacy. In [?], RSSI-based distance estimation at fog nodes is combined with blockchain-stored position proofs for trajectory-based Sybil identification in vehicular fog networks. In [?], a multi-level blockchain trust mechanism (BMLT-SA) combines V2V machine learning at the vehicle tier with RSU verification and blockchain for Sybil detection. While these approaches provide strong non-repudiation, they introduce confirmation latency incompatible with real-time vehicular constraints and do not address federated or AI-driven detection in SDVEN architectures.

Graph and Temporal Sequence Based Detection

Recent works model vehicular communication as a graph to capture relational Sybil attack patterns. In [?], a deep learning mechanism identifies Sybil nodes by detecting associativity between beacon senders using dynamic time warping (DTW) similarity on common vehicle characteristics including Euclidean distance, speed, and flow. In [?], a temporal heterogeneous graph attention network (THGAT) is proposed for joint Sybil detection and attacker tracing in VANETs, constructing six attack-behavior features and applying both network-layer and physical-layer tracing using the VeReMi-Extension dataset. In [?], the challenge of detecting Sybil attacks in cooperative ITS environments is analyzed, identifying

Table 2.1: Comparison of Related Sybil Attack Detection Works

Work	Domain	FL	LLM	Multi-Agent	Mobility-Aware	Control-Plane
Yao et al. [?]]	VANET	✗	✗	✗	Partial	✗
Tulay & Koksal [?]]	VANET	✗	✗	✗	✓	✗
Chen et al. [?]]	VANET	✗	✗	✗	Partial	✗
Sultana et al. [?]]	VANET	✗	✗	✗	✓	✗
Baza et al. [?]]	VANET	✗	✗	✗	✗	✗
Benadla et al. [?]]	VFN	✗	✗	✗	✗	✗
Nayak et al. [?]]	SDVN	✗	✗	✗	✗	Partial
Azam et al. [?]]	VANET	✗	✗	✗	✗	✗
Hamhoum et al. [?]]	VANET	✗	✓	✗	✗	✗
Khan et al. [?]]	VANET	✗	Partial	✗	✗	✗
Muhammad [?]]	VANET	✓	✗	✗	✗	✗
Naik & Dondeti [?]]	VANET	✓	✗	✗	✗	✗
Asad & Otoum [?]]	SDVEN	✓	✗	✗	✗	✗
Jai Vinita et al. [?]]	SDVF	✓	✗	✗	✗	✗
Altmemi et al. [?]]	VSN	Partial	✗	✗	✗	✗
Proposed	SDVEN	✓	✓	✓	✓	✓

VFN: Vehicular Fog Network; VSN: Vehicular Social Network; PQC: Post-Quantum

Cryptography; Dual-Mode: lightweight rule-based + full cryptographic detection/mitigation.

that pseudonym-change strategies and message silence techniques substantially increase Sybil stealthiness, motivating contextual reasoning beyond feature-level detection.

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2.2 Gaps in Literature

2.2.1 Limitations of Existing Approaches

Despite significant progress in Sybil attack detection for vehicular and software-defined vehicular networks, existing approaches exhibit several critical limitations. Existing traditional, machine learning, deep learning, and federated learning-based approaches predominantly rely on strong trust assumptions, particularly considering RSUs and SDN controllers as inherently trustworthy. This assumption introduces a single point of failure, as compromised or Byzantine control-plane entities can manipulate detection outcomes, disrupt global model aggregation, or facilitate coordinated insider attacks. However, the realistic threat of malicious control-plane components has been largely overlooked in current literature [11],[27],[28],[29],[30].

Furthermore, most existing detection frameworks adopt unidirectional or partially decentralized monitoring, where trust evaluation is confined to vehicle-level observations or centralized aggregation at RSUs or controllers. Such designs fail to capture complex cross-layer attack scenarios involving interactions among vehicles, RSUs, and controllers. The absence of mutual and cross-entity trust monitoring mechanisms limits the robustness of these systems against multi-vector Sybil attacks [28].

Although federated learning has emerged as a promising solution for privacy-preserving Sybil detection, current FL-based approaches predominantly employ task-specific conventional machine learning or lightweight DL models. These models lack advanced reasoning, contextual awareness, and semantic understanding, which are essential for detecting complex and adaptive attack behaviors in highly dynamic SDVEN environments. Moreover, existing FL frameworks largely assume honest aggregators and do not use trust-aware or adversarial-resilient model coordination methods [27],[28],[29],[30].

The application of LLMs in vehicular network security remains in its early stages, with prior studies primarily focusing on transformer-based intrusion detection or explainable enhancements [23],[24].

2.2.2 Research Gap

The following specific, distinct gaps are identified from the preceding review:

G1 – Mobility-Aware Threat Modeling: No existing SDVN/SDVEN work [? ? ?] explicitly formulates how vehicle mobility (topology churn, RSU handoff exploitation, short contact windows, fading-induced RSSI ambiguity) amplifies Sybil attack impact at each network tier.

G2 – Untrusted Control-Plane Entities: All existing FL-based detection frameworks [? ? ? ?] assume RSUs and SDN controllers are honest aggregators. No prior work models or detects Sybil attacks originating from compromised RSUs or SDN controllers.

G3 – No Federated LLM Multi-Agent Detection: LLM application to vehicular security is limited to centralized, non-federated inference [? ? ?]. No work deploys federated LLM agents at the RSU edge tier for Sybil-specific contextual reasoning.

G4 – No Dual-Mode Lightweight and Full Detection: Existing works propose either computationally heavy models or lightweight classifiers. No framework simultaneously offers lightweight rule-based signature detection and a full cryptographically enforced mitigation path, limiting practical deployment feasibility.

G5 – No Post-Quantum Cryptographic Mitigation: Existing authentication protocols use classical ECC [?] or conventional PKI. No Sybil detection framework for SDVEN integrates post-quantum primitives (e.g., CRYSTALS-Dilithium, Kyber) for long-term mitigation security.

These five gaps collectively define the design space addressed by this work.

Chapter 3

Methodology

This project focuses on implementing a research design, combining simulation-based experimentation and iterative optimization to develop and validate the federated LLM-based multi-agent system for Sybil attack detection in SDVEN. The iterative framework will be consisted with exploratory phase for attack modeling, development phase for system prototyping, optimization and validation phase for performance tuning and evaluation phase for outcome analysis. This approach ensures adaptability to Sybil attacks while prioritizing resource efficiency and data privacy.

3.1 Threat Model

The threat model exclusively focuses on Sybil attacks that creates fake identities in the network using malicious vehicle node,RSU or SDN controller with in the three tier SDVEN architecture.

3.1.1 Network Assumptions

1. SDVEN network provides reliable coverage of RSUs and SDN controller availability.
2. Vehicle On-Board Unit (OBU)s having lightweight resources that can support small ML model training and RSUs having better resources that could able to run LLM agent functions.
3. Secure communication channels are deployed between network tiers for token exchange FL flows.
4. Attackers cannot physically manipulate vehicle Vehicle Identification Number (VIN)s and exceed the realistic vehicle constraints.

3.1.2 Attacker Capabilities and Assumptions

1. The attacker is assumed to having complete knowledge of the network architecture of the system.

2. The attacker is capable of forging multiple fake vehicle identities and communicate with other nodes and controllers appearing as multiple vehicle nodes.
3. When the attacker is a vehicle node it cannot exceed the realistic vehicle mobility behaviour and realistic RSSI patterns.
4. The attacker is assumed to cannot launch attack on physical layer and controller layer of SDVEN architecture concurrently. Sybil attacks can be initiated at different SDVEN network layers using distinct approaches

(a) Vehicle Tier Attacks

Malicious vehicle who is an insider or outsider node of the network, creates multiple Sybil nodes and they interact with RSU directly or indirectly through the other vehicle nodes by appearing simultaneously or non-simultaneously over the time.

(b) RSU Tier Attacks

Malicious RSU adds multiple Sybil nodes into the network, approves the Sybil nodes and shares the node ids with SDN controller and vehicular nodes with faking them as legitimate vehicle nodes.

(c) SDN Tier Attacks

Malicious SDN controller adds multiple Sybil nodes into the network and registers them into the network and shares the node ids with RSUs and vehicular nodes with faking them as legitimate vehicle nodes. Poisoning FL aggregations to manipulate the Sybil attack detection models

3.1.3 Identified Sybil Attack Variants

When implementing the system, six main attack variants are focused, each showing a unique behaviour in SDVEN environment.

1. Outsider Sybil Attacks

Step 1. A malicious vehicle that not registered and authenticated in the network broadcasts packets that contains Sybil IDs.

Step 2. The near vehicle nodes and RSUs can receive that packets and thinks those nodes are legitimate nodes.

Step 3. If the Sybil IDs could not be identified by legitimate nodes or RSU, they may forward to SDN controllers.

Step 4. SDN controller registers the received Sybil IDs as legitimate vehicle nodes and malicious node can able to manipulate the network with Sybil nodes.

2. Insider Direct Simultaneous Sybil Attacks

Step 1. As shown in Figure 3.1 a malicious vehicle is already registered into the network and that can authenticate through the RSUs broadcasts packets that contains Sybil IDs to nearby RSUs and legitimate nodes simultaneously.

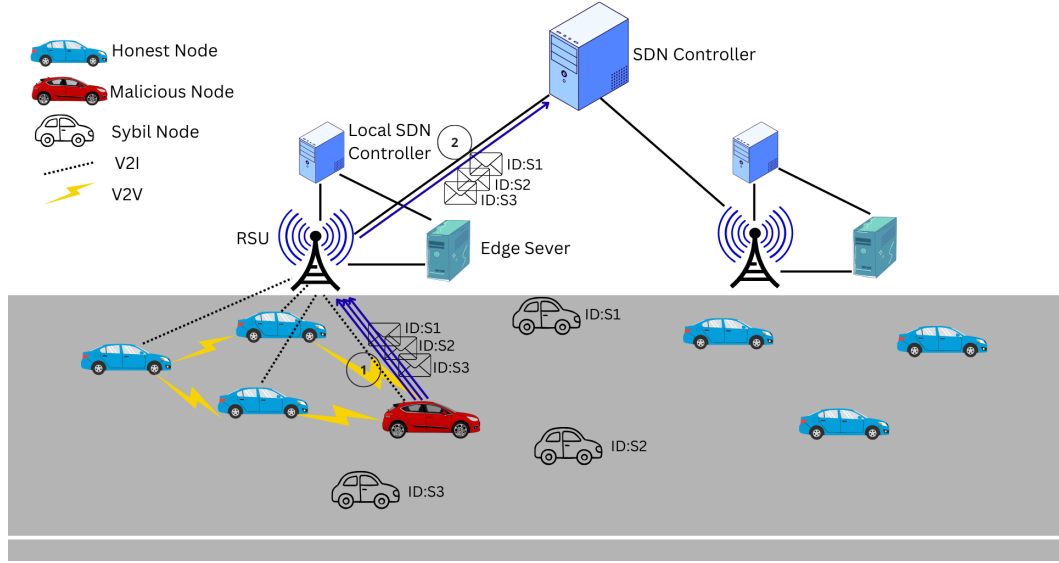


Figure 3.1: Packet Flow Diagram of Insider Direct Simultaneous Sybil Attack

Step 2. If the RSU could not identify the Sybil IDs, accept them as real nodes.

Step 3. Then with the RSU packets including Sybil IDs sends into SDN controller and they deceive also the SDN controller.

Step 4. SDN controller registers the received Sybil IDs as legitimate vehicle cluster in the network and malicious node can able to manipulate the network with Sybil nodes.

3. Insider Direct Non-Simultaneous Sybil Attacks

Step 1. Figure 3.2 shows that a malicious vehicle is already registered into the network and it appears as legitimate node and broadcasts packets including set of Sybil IDs at time t_1 .

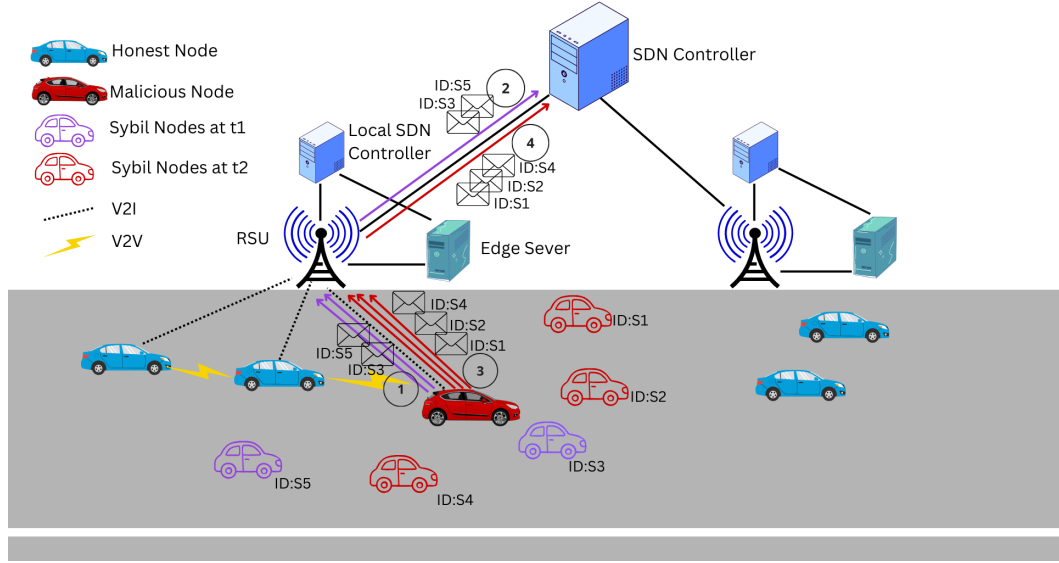


Figure 3.2: Packet Flow Diagram of Insider Direct Non-Simultaneous Sybil Attack

Step 2. If the RSU unable to identify the Sybil IDs, it sends packets including Sybil IDs to the SDN controller. Then SDN controller thinks the received IDs are legitimate and register them in the network.

Step 3. At time t_2 malicious vehicle broadcasts packets including another set of Sybil IDs.

Step 4. If the acRSU fails to identify the Sybil IDs, repeats the Step 2. SDN controller deceives the Sybil IDs as legitimate vehicles appear time to time in the network.

4. Insider Indirect Sybil Attacks

Step 1. As shown in Figure 3.3, a vehicle node already registered sends packets including Sybil IDs to the nearby legitimate nodes.

3.1.4 Mobility-Induced Sybil Threat Amplification in SDVEN

Vehicle mobility introduces four structural vulnerabilities that uniquely amplify Sybil exploitation in SDVEN beyond static network threat models.

Topology Churn: Let the neighbor set of vehicle v_i at time t be $\mathcal{N}_i(t)$. Under vehicular mobility with speed s , the neighbor set evolves as $\mathcal{N}_i(t + \delta) \neq \mathcal{N}_i(t)$, with churn rate $\rho_c = \frac{|\mathcal{N}_i(t) \triangle \mathcal{N}_i(t+\delta)|}{|\mathcal{N}_i(t)|}$, where $\triangle$ denotes symmetric difference. High ρ_c normalizes the appearance of new identities, making Sybil node injection indistinguishable from legitimate arrivals.

RSU Handoff Exploitation: During RSU coverage transition, an attacker can re-register under a new pseudonym ID' at the target RSU r_j before revocation of the old identity ID propagates from RSU r_i to the SDN controller, exploiting the synchronization gap Δt_{sync} defined as:

$$\Delta t_{\text{sync}} = t_{\text{register}}(r_j) - t_{\text{revoke}}(r_i \rightarrow \text{SDN}) \quad (3.1)$$

When $\Delta t_{\text{sync}} > 0$, duplicate identities coexist in the network.

Short Contact Windows: At relative speed v_{rel} with communication range R , the contact duration $T_c = \frac{2R}{v_{\text{rel}}}$. At highway speeds ($v_{\text{rel}} = 240$ km/h, $R = 300$ m), $T_c \approx 9$ s, which constrains multi-round identity verification protocols. This forces single-pass detection mechanisms at the RSU tier.

RSSI Fading Under Mobility: Received signal strength at receiver r from transmitter t under vehicular fading is modeled as:

$$P_r = P_t - 10\alpha \log_{10}(d) + X_\sigma + X_f(v) \quad (3.2)$$

where α is the path loss exponent, $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ models shadowing, and $X_f(v)$ models velocity-dependent fast fading. The mobility-induced variance $X_f(v)$ overlaps with RSSI differentials between co-located Sybil nodes, degrading RSSI-only detection.

Together, these factors motivate a multi-feature, SDN-aware, and edge-distributed detection architecture rather than single-node or RSSI-only approaches.

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3.1.5 Attack Signatures of Sybil Variants

Table 3.1 formalizes the observable attack signatures for each Sybil variant. Let $\mathcal{B}_i(t)$ denote the beacon set transmitted by node i at time t , RSSI_{ij} denote the received signal strength from node i at observer j , $\text{loc}_i(t)$ denote the claimed location of node i , and τ_i denote the registration token of node i .

The RSSI co-location threshold for simultaneous attacks is derived from the fading model in (??). For the simultaneous variant, two distinct IDs i and j are flagged as co-located if:

$$\Phi_{\text{coloc}}(i, j) = \exp\left(-\frac{(\text{RSSI}_{ik} - \text{RSSI}_{jk})^2}{2\sigma_{\text{ch}}^2}\right) > \gamma_{\text{co}}, \quad \forall k \in \mathcal{O} \quad (3.3)$$

where σ_{ch}^2 is the channel variance estimated from the mobility fading model and $\mathcal{O}$ is the set of observer nodes.

Table 3.1: Formal Attack Signatures for Six Sybil Variants in SDVEN

Attack Variant	Primary Signature	Formal Indicator	Detection Tier
Outsider Sybil	Missing or invalid token τ	$\tau_i \notin \mathcal{T}_{\text{valid}}$, where $\mathcal{T}_{\text{valid}}$ is the RSU-cached token registry	RSU (token check)
Insider Direct Simultaneous	Multiple IDs with correlated RSSI from same physical origin	$\exists i \neq j : \ \text{RSSI}_{ik}(t) - \text{RSSI}_{jk}(t)\ < \epsilon_{\text{RSSI}}, \forall k \in \mathcal{O}$	Vehicle / RSU
Insider Direct Non-Simultaneous	Temporal burst of new IDs from single spatial region	$\Delta n_{\text{new}}(\Delta t) > \theta_n$ and $\text{var}(\text{IAT}_i) > \theta_{\text{IAT}}$	RSU / SDN
Insider Indirect	Forwarded Sybil IDs with correlated motion to legitimate relay	$\text{sim}(\text{traj}_i, \text{traj}_j) > \theta_{\text{traj}}$ and $\text{src}(m_j) = v_k$ (legitimate)	Vehicle / RSU
Malicious RSU	RSU approves IDs without valid vehicle-tier RSSI confirmation	$\text{approve}(r, \text{ID}) = 1$ and $\nexists k : \text{RSSI}_{r, \text{ID}} \in \mathcal{V}_k$	SDN / Vehicle
Malicious SDN Controller	Global FL model diverges; controller issues tokens not corroborated by RSUs	$\ \mathbf{w}_{\text{global}} - \text{median}(\mathbf{w}_{\text{RSU}})\ _2 > \theta_{\text{div}}$ and $\nexists r : \text{corroborate}(\text{token}, r)$	RSU collective

The temporal burst signature for the non-simultaneous variant is quantified by the inter-arrival time (IAT) entropy:

$$H_{\text{IAT}}(i) = - \sum_k p_k \log_2 p_k, \quad \text{flag if } H_{\text{IAT}}(i) < \theta_H \quad (3.4)$$

where p_k is the empirical probability of IAT bin k . Abnormally low entropy indicates bursty, non-organic identity injection.

The indirect attack trajectory similarity is measured by the normalized dynamic time warping distance:

$$\text{sim}_{\text{DTW}}(i, j) = 1 - \frac{\text{DTW}(\text{traj}_i, \text{traj}_j)}{\max(\text{DTW})} \quad (3.5)$$

A value exceeding threshold θ_{traj} indicates that node j is spatially shadowing node i , consistent with indirect Sybil relaying.

The malicious SDN controller signature uses model parameter divergence:

$$\Delta_{\text{div}} = \left\| \mathbf{w}_{\text{SDN}} - \text{median}_{r \in \mathcal{R}} \mathbf{w}_r \right\|_2 \quad (3.6)$$

where $\mathbf{w}_{\text{SDN}}$ is the global model broadcast by the controller and $\mathbf{w}_r$ is the locally aggregated model at RSU r . If $\Delta_{\text{div}} > \theta_{\text{div}}$, the controller is flagged.

3.2 System Architecture

As shown in Figure 3.6, the proposed solution is designed as a three-tier FL architecture incorporating LLM-based multi-agent coordination.

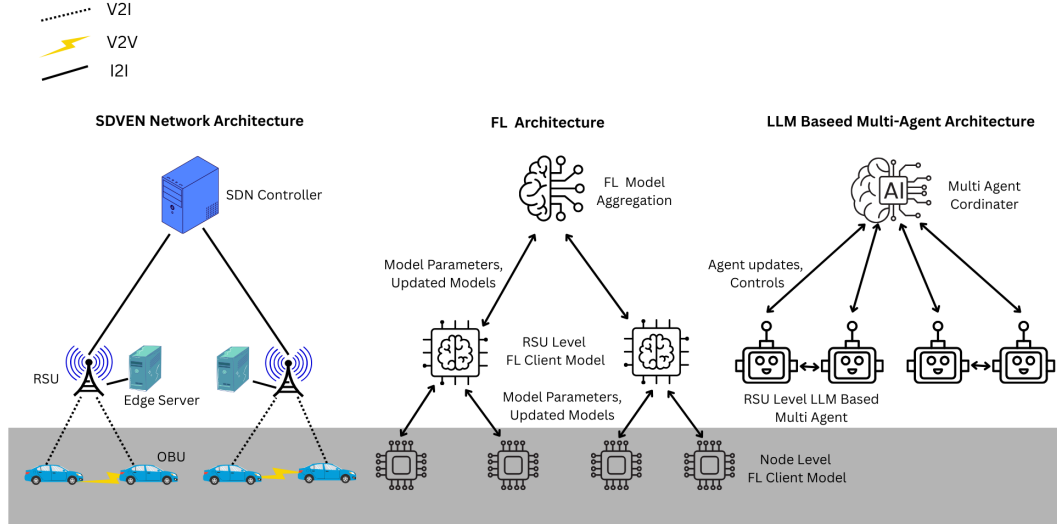


Figure 3.6: High level System Architecture Diagram

The system architecture is based on SDVEN network architecture that contains vehicular nodes, RSU with edge servers and SDN controller. The system functional modules those shown in Figure 3.7 are designed to be align with the three tier architecture of the system shown in Figure 3.6

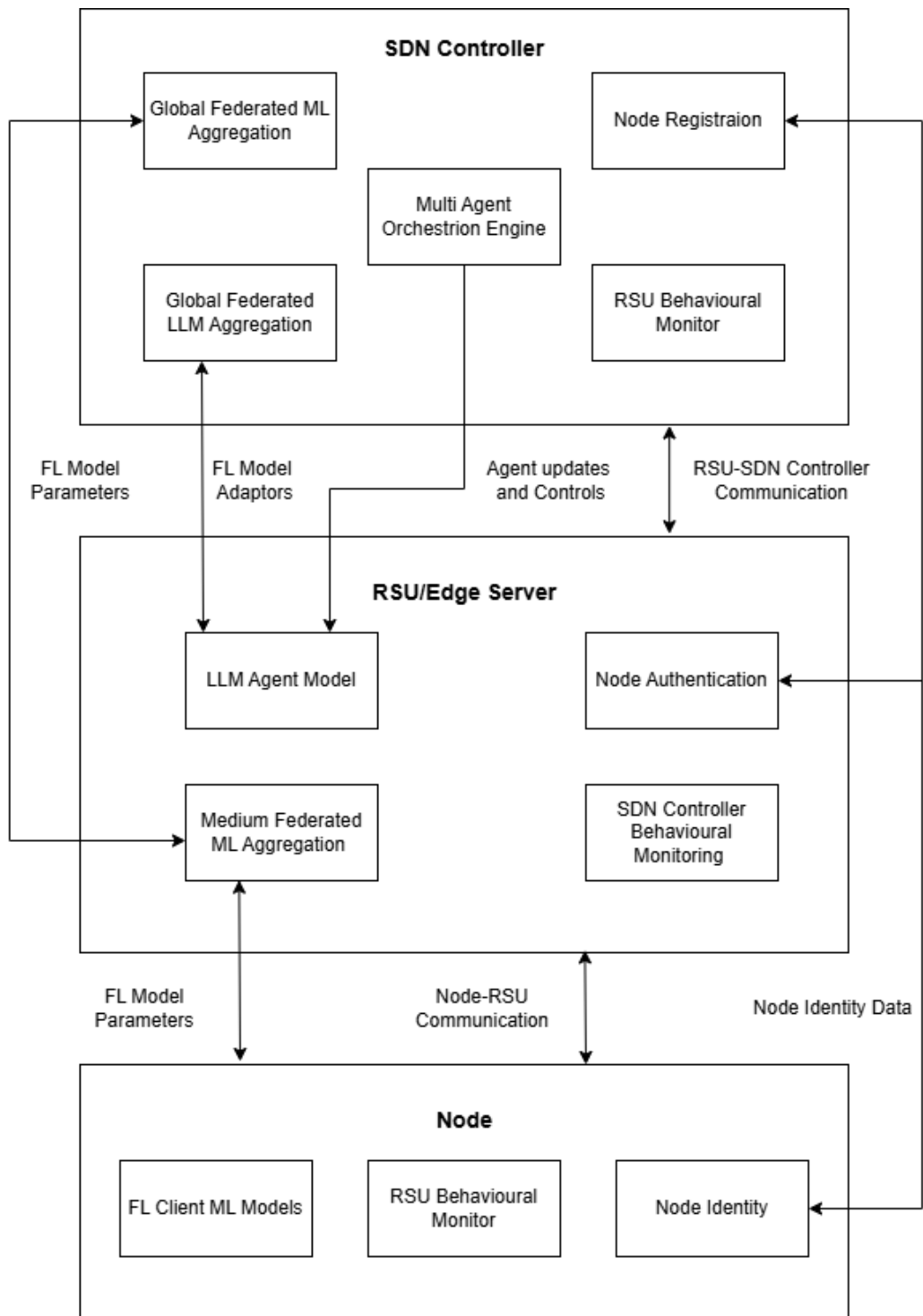


Figure 3.7: System Functional Block Diagram

1. Vehicle Node Tier

Vehicular Node Tier is implemented on OBUs in the SDVEN those have recourse constrain. Therefore, only light weight FL client modules are deployed in this tier. FL client modules in vehicle node tier are ML models those are trained for evaluate the vehicular nodes based in the recovering data from the other nodes. RSSI analyzer, temporal data analyzer and trust score analyzer will be the ML based models in the proposed system. The node level FL client nodes are training and sharing the gradients with parent models and receive the model updates from the global models.

2. RSU/Edge Server Tier

In SDVEN RSU is connected with edge server that have more resources than vehicle node. Light weight fine-tuned LLM based multi agents are deployed in RSU edge servers. The agents will work on analyzing the contextual data and outputs of the ML based multi analyzers those are RSSI analyzer, temporal data analyzer and trust score analyzer. The multi agents will output decisions on the nodes with reasons and recommending actions to the identified Sybil nodes. The mid level aggregation models of the ml analyzer models will deploy on the RSU

3. SDN Controller Tier

SDN controller is the top level of SDVEN that works with the highest resources in the architecture. In the proposed architecture Global FL ML model aggregation and Federated LLM adapter aggregation are deployed in SDN controller. They manage and updated the client FL models in the RSU edge servers and vehicular nodes. Multi-Agent orchestration engine mangers and control the LLM based agents in the RSU level from SDN controller. Central authority on LLM, ML based FL models and LLM multi-agents are integrated in SDN controller Tier

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3.2.1 Cryptographic Placement Analysis

A critical design question is whether cryptographic mechanisms alone completely mitigate Sybil attacks, or whether they must be combined with AI-based detection. Table 3.2 analyzes the mitigation coverage of cryptography against each attack variant.

The analysis reveals that cryptography *completely* mitigates only v_1 (Outsider attacks), where token-based Kyber KEM authentication prevents all unauthorized network entry. For all remaining variants (v_2-v_6), a legitimate cryptographic credential is held by the attacker, rendering cryptography alone insufficient. Consequently:

- **Post-detection cryptography** is applied for v_1 , v_3 , and v_4 : AI detection identifies the Sybil behavior; threshold Dilithium revocation and Kyber re-keying then terminate the attack.

Table 3.2: Cryptographic Mitigation Coverage per Sybil Attack Variant

Variant	Cryptographic Mechanism	Complete Mitigation?	Placement
v_1 : Outsider	Token-based authentication (Kyber KEM)	Yes	Post-detection
v_2 : Insider Direct Simultaneous	PKI + token binding cannot prevent a legitimate node creating concurrent Sybil IDs from one device	No	Pre + Post
v_3 : Insider Direct Non-Simultaneous	Token refresh prevents reuse but not sequential identity cycling within token lifetime	No	Post-detection
v_4 : Insider Indirect	Cryptographic message integrity cannot detect relay-forwarded Sybil IDs that hold valid tokens	No	Post-detection
v_5 : Malicious RSU	Single RSU cryptographic authority is itself compromised; threshold Dilithium (t -of- n) prevents unilateral forgery	Partial	Pre + Post
v_6 : Malicious SDN	SDN controller signs tokens; compromise nullifies single-authority PKI; MPC aggregation limits model poisoning	Partial	Pre + Post

- **Pre-detection cryptography** is applied for v_2 , v_5 , and v_6 : threshold Dilithium (t -of- n) co-authorization at the RSU tier prevents unilateral Sybil

approval *before* identity injection succeeds, reducing the attack surface that AI detection must cover, and thereby improving detection precision.

This placement design directly follows the supervisor’s principle: cryptography is applied post-detection when it fully resolves the attack, and pre-detection when it only partially mitigates, to improve conditions for AI-based detection.

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3.2.2 Three-Tier Logical Key Hierarchy and Key Establishment

Inspired by Kyber-based key encapsulation in PQC vehicular authentication protocols [? ?], a three-level Logical Key Hierarchy (LKH) governs session key distribution across SDVEN tiers prior to detection operation.

Hierarchy Structure:

$$\mathcal{K} = \{K_{root}, \{K_{zone}^{(r)}\}_{r \in \mathcal{R}}, \{K_{sess}^{(v)}\}_{v \in \mathcal{V}}\} \quad (3.7)$$

where K_{root} is the SDN controller master key, $K_{zone}^{(r)}$ is the zone key for RSU r , and $K_{sess}^{(v)}$ is the session key for vehicle v .

Zone Key Encapsulation (SDN $\rightarrow$ RSU):

$$(ct_{zone}^{(r)}, K_{zone}^{(r)}) \leftarrow \text{ML-KEM.Enc}(\mathbf{pk}_r, K_{root} \parallel \text{ID}_r \parallel ts) \quad (3.8)$$

Session Key Encapsulation (RSU $\rightarrow$ Vehicle):

$$(ct_{sess}^{(v)}, K_{sess}^{(v)}) \leftarrow \text{ML-KEM.Enc}(\mathbf{pk}_v, K_{zone}^{(r)} \parallel \text{ID}_v \parallel ts) \quad (3.9)$$

Key Derivation for Detection Channel: Feature vectors transmitted by vehicle v to RSU r for trust scoring and signature computation are authenticated using a detection channel key derived from the session key:

$$K_{det}^{(v)} = \text{HKDF}(K_{sess}^{(v)}, \text{"detection"} \parallel \text{ID}_v \parallel ts) \quad (3.10)$$

where HKDF is HMAC-based key derivation. Upon Sybil revocation, $K_{sess}^{(v)}$ is replaced via Eqs. (??)–(??), propagating updated keys downward through the hierarchy per Eqs. (3.8)–(3.9).

3.2.3 Dual Federated Learning Framework

The proposed solution employs a dual FL framework that is suitable in the three-tier hierarchy of SDVEN networks to detect and prevent Sybil attacks while preserving vehicle data privacy. This dual FL architecture made with separate FL flows as ML-FL for conventional ML models and LLM-FL for fine-tuned LLM based agents.

1. ML-FL flow

ML-FL operates across all three tiers in the solution architecture with optimized training cycles. Lightweight FL clients are executed in vehicle nodes

that process RSSI and data similarity ML models. They upload the encrypted model gradients to associate RSUs and edge servers connected to RSUs perform local aggregation of these gradients. Edge computers also employees their FL client models on RSSI analyzer, temporal data analyzer and trust score analyzer ML based models and share the gradients with SDN controller. Global FL ML model aggregation is done in SDN controllers using the shared gradients and produce global ML model parameters, which are distributed again into FL clients. This ML-FL flow ensures the data privacy on each tier in the hierarchy and optimize the resources of each layer.

2. LLM-FL Flow

LLM-FL is connection between the LLM agentic models in edge servers and LLM-FL aggregators in SDN controllers. RSU LLM agents are based on fine-tuned light weight LLM models. When they are fine tuning instead of transmitting full LLM weights, RSU hosted models upload only quantized LoRA adapters. Federated LLM adapter aggregator produces globally optimized LLM adapters and the updated adapters synchronize back to RSU hosted LLMs. This LLM-FL flow enabling network wide consistency in LLM based analysers while maintaining the computational feasibility at edge servers.

LLM Agent Architecture and Fine-Tuning Design

Three specialized agents operate at the RSU edge tier: **Agent** a_1 (Message Pattern Detection Agent) analyzes beacon frequency, entropy, and inter-arrival irregularities; **Agent** a_2 (Trust Scoring Agent) integrates multi-factor trust sub-scores with historical behavioral context; **Agent** a_3 (Temporal Data Analyzer Agent) processes RSSI time series and trajectory continuity features. Each agent is a fine-tuned lightweight LLM (e.g., Mistral-7B or LLaMA-3) adapted using Low-Rank Adaptation (LoRA) [?].

LoRA Adapter Formulation: For each weight matrix $W_0 \in \mathbb{R}^{d \times k}$ of the pre-trained LLM, the adapted weight is:

$$W = W_0 + \Delta W = W_0 + B^{(a)} A^{(a)} \quad (3.11)$$

where $B^{(a)} \in \mathbb{R}^{d \times r}$, $A^{(a)} \in \mathbb{R}^{r \times k}$, $\text{rank } r \ll \min(d, k)$, and W_0 is frozen during fine-tuning. Only $\{B^{(a)}, A^{(a)}\}$ are updated and shared via LLM-FL, reducing communication to $\mathcal{O}(r(d + k))$ parameters per round.

Fine-Tuning Objective: Each agent a is fine-tuned on locally generated SDVEN simulation data $\mathcal{D}_a = \{(\mathbf{c}_i, y_i^{(k)})\}$ minimizing the cross-entropy loss over attack variant labels:

$$\mathcal{L}_{LLM}^{(a)} = -\frac{1}{|\mathcal{D}_a|} \sum_{(\mathbf{c}_i, y_i) \in \mathcal{D}_a} \sum_{k=0}^6 y_i^{(k)} \log \mathcal{P}_{LLM}^{(a)}(y^{(k)} | \mathbf{c}_i) \quad (3.12)$$

where $y^{(0)}$ is the legitimate class and $y^{(k)}$, $k \in \{1, \dots, 6\}$, are the six Sybil attack variant classes.

Context Window Structure: The input context $\mathbf{c}_i$ fed to each agent is a

structured token sequence:

$$\mathbf{c}_i = \left[\underbrace{f_{rssi}, f_{temp}, f_{beacon}, f_{token}}_{\text{feature tokens}} \mid \underbrace{\mathcal{S}_1, \dots, \mathcal{S}_6}_{\text{signature tokens}} \mid \underbrace{\mathcal{T}(i, t)}_{\text{trust token}} \mid \underbrace{v_{rel}, \rho_c, \Delta t_{sync}}_{\text{mobility tokens}} \right] \quad (3.13)$$

The inclusion of mobility tokens $\{v_{rel}, \rho_c, \Delta t_{sync}\}$ from the threat model (Section ??) enables agents to condition Sybil classification on the current mobility regime, directly addressing Gap G1.

LLM-FL LoRA Adapter Aggregation: At the SDN controller, quantized adapter updates from all RSU agents are aggregated:

$$\bar{B}^{(a)}, \bar{A}^{(a)} = \frac{\sum_{r \in \mathcal{R}} |\mathcal{D}_r| \left(B_r^{(a)}, A_r^{(a)} \right)}{\sum_{r \in \mathcal{R}} |\mathcal{D}_r|} \quad (3.14)$$

Global adapters $\bar{B}^{(a)}, \bar{A}^{(a)}$ are redistributed to all RSU agents, preserving network-wide LLM consistency without transmitting full model weights.

...

FL Convergence Under Non-IID Vehicular Data

Vehicular data is inherently non-IID across RSU coverage zones due to heterogeneous traffic densities and mobility patterns. Under the FedProx objective (Eq. (??)) with proximal coefficient $\mu > 0$, the global model converges at round T with bound:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla \mathcal{L}(\theta_g^{(t)})\|^2] \leq \frac{2(\mathcal{L}(\theta_g^{(0)}) - \mathcal{L}^*)}{T \eta_g} + \frac{\eta_g L^2 \sigma^2}{B} + \Gamma_{nonIID} \quad (3.15)$$

where η_g is the global learning rate, L is the gradient Lipschitz constant, σ^2 is the gradient variance, B is the local batch size, and Γ_{nonIID} is the non-IID divergence term bounded by the degree of data heterogeneity across RSU zones. The proximal term $\mu \|\theta_v - \theta_g\|^2$ in Eq. (??) controls Γ_{nonIID} by limiting local model drift, with larger μ reducing divergence at the cost of slower convergence. Under MPC-secured aggregation (Eqs. (??)–(??)), Byzantine RSU gradient contributions are bounded by the (t, n) -threshold, ensuring at most $n - t$ poisoned gradients enter aggregation.

3.2.4 Identity Management and Authentication System

To counter outsider Sybil attacks where the external nodes to the network forge fake vehicle identities the proposed solution introduces a light weight identity management system leveraging the SDVEN architecture. This identity management process can divide into main two processes.

1. Global Registration Process When a vehicle entering into the network registration manager function register the node using Vehicle Identification Number, GPS coordinated and timestamp proof. The SDN controller generates

and issues a short lived cryptographic token bond uniquely to the vehicle. This Registration process occurs once per token lifetime that minimizes the SDN controller identity ownership.

2. Local Authentication Process

During the data transmissions from the vehicle node every beacon transmission with RSU identity token validates with token cached token registry that synchronized with the global identity registration process. This process will eliminate the ability to create Sybil nodes by external vehicles.

3.2.5 Decentralized Cross-Tier Monitoring System

The proposed architecture introduces a comprehensive decentralized monitoring system that monitors the other tiers from each tier.

1. RSU Monitoring from Vehicle Node
2. Vehicle Node Monitoring from RSU
3. SDN Controller Monitoring from RSU
4. RSU Monitoring SDN from Controller

Using the trust score and RSSI data based analyzing functions are proposed to use in Monitoring system. It is developed using existing models in each tier and evaluating their behaviour with Sybil node detection.

3.2.6 Sybil Attack Detection Mechanisms

The proposed three-tier architecture is designed to addresses all major Sybil attack variants through the specialized mechanisms in appropriate networks layers by leveraging SDVEN network layout. Fundamental information security mechanisms such as digital signatures for validation, hashing to protect transferred data and encryption across plane communication serve as the foundation layer for reliability. The specific attack detection techniques based on different characteristics are build upon these foundation layer.

1. Outsider Sybil Attacks

Outsider Sybil attacks are launched by external node trying to unauthorized network access or RSU communication. The proposed identity management system with token-based authentication is detected outsider attacks, if a node failed with local validation, it is isolated and blocking the transactions through the node.

2. Insider Direct Simultaneous Sybil Attacks

Insider direct simultaneous Sybil attacks happens when a malicious vehicle generates Sybil nodes and appearing in the network concurrently. The RSSI based node classifier will be used to identify the Sybil IDs those are showing similar RSSI values with different location data. Trust score will calculate using factors such beacon rate and entropy will also work to detect this type of Sybil IDs.

3. Insider Direct Non-Simultaneous Sybil Attacks

Insider direct non-simultaneous Sybil attacks happens when a malicious vehicle generates Sybil nodes and appearing in the network time to time. Temporal data analyzer model will identify unnatural patterns such as sudden bursts of new IDs from the same source, irregular inter-arrival times, IDs that “appear/disappear” without realistic movement. Trust score will also calculate considering the rate of node injecting new IDs to the system and beacon rate and entropy.

4. Insider Indirect Sybil Attacks

In insider indirect Sybil attacks, Legitimate nodes are used to transactions of Sybil attacks to RSU and other nodes. FL-client models that deployed in vehicle nodes are analyzing receiving data from other vehicular networks. Using RSSI based node classifier observes the behavioural patterns of the other nodes and classify them. The trust score will consider the beacon rate and entropy of the node.

5. Malicious RSU Sybil Attacks

Malicious RSU Sybil Attacks happen in the controller plane when a RSU become malicious and starting to approve fake identities, providing fake identities and block legitimate traffic. RSU monitoring function from nodes and SDN controllers observing the RSU behaviour with Sybil nodes. If RSU approves Sybil nodes continuously or labels legitimate nodes as Sybil, using trust score of RSU and monitoring data from other two tiers detect the malicious RSUs.

6. Malicious SDN Controller Sybil Attacks

Malicious SDN controller works to manipulate global FL aggregations and issue tokens to Sybil nodes because the centralized authority to operate the SDVEN network. Using RSU FL clients and trust of authentication process, Trust score of SDN controller can be detected. Considering the multiple RSUs trust on SDN controller take the decision about the SDN controller behaviour.

In our proposed solution global LLM model will be deployed in SDN controller level and LLM agents will be in RSU level. LLM based agents will worked on aggregating contextual meta data and results of RSSI based analyzer, trust score analyzer and temporal data analyzer models. It will provide a decision on the node with reasoning that will identify the specific attack type and recommend an appropriate action such as immediate packet drop, temporary monitoring, exclusion from SDN registration, or propagation of a blacklist alert to the network.

3.2.7 Per variant detection logic

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3.4 Dual-Mode Detection and Mitigation Framework

The proposed framework operates in two modes selected by a resource-awareness function to balance detection performance against deployment feasibility. Let ρ_v denote the residual processing capacity of a vehicle node, ρ_r denote the RSU edge server load, and Λ denote the current network threat level derived from the cross-tier monitoring system.

Algorithm 1 Per-Variant Detection Logic — Vehicle-Tier Attacks (v_1-v_4).

```

1: procedure DETECTVEHICLETIER( $i, t, \mathcal{M}$ )
2:   //  $v_1$ : Outsider
3:   if  $\tau_i \notin \mathcal{R}_{SDN}(t)$  then
4:     return  $v_1$ , MITIGATE( $i, v_1, \mathcal{M}$ )
5:   end if
6:   //  $v_2$ : Insider Direct Simultaneous
7:   Compute  $\mathcal{S}_2(i, t)$  per Eq. (??)
8:   if  $\mathcal{S}_2(i, t) > \theta_2^L$  and  $|\{j : \rho_{ij} > \theta_\rho\}| > 1$  then
9:     return  $v_2$ , MITIGATE( $i, v_2, \mathcal{M}$ )
10:  end if
11:  //  $v_3$ : Insider Direct Non-Simultaneous
12:  Compute  $\mathcal{S}_3(i, k)$  per Eq. (??) for most recent sequential ID pair
13:  if  $\mathcal{S}_3(i, k) > \theta_3^L$  then
14:    return  $v_3$ , MITIGATE( $i, v_3, \mathcal{M}$ )
15:  end if
16:  //  $v_4$ : Insider Indirect
17:  Compute  $\mathcal{S}_4(i)$  per Eq. (??)
18:  if  $\mathcal{S}_4(i) > \theta_4^L$  then
19:    return  $v_4$ , MITIGATE( $i, v_4, \mathcal{M}$ )
20:  end if
21:  return legitimate
22: end procedure

23: function MITIGATE( $i, v_k, \mathcal{M}$ )
24:   if  $\mathcal{M} = \mathcal{M}_L$  then
25:     Drop all packets from  $i$ ; blacklist  $\tau_i$  locally
26:   else
27:     THRESHREVOKE( $i, t, \mathcal{S}_t, t_{thr}, \mathbf{pk}_{new}$ ) ▷ Alg. ??
28:     IPFSREVOCATE( $i, t, \mathcal{M}, \mathcal{R}_{SDN}$ ) ▷ Alg. ??
29:   end if
30: end function

```

3.4.1 Mode Selection

The dual-mode selector $\mathcal{M}$ is defined as:

$$\mathcal{M} = \begin{cases} \text{LIGHTWEIGHT} & \text{if } \rho_v < \rho_{\min} \text{ or } \Lambda < \Lambda_{\text{lo}} \\ \text{FULL} & \text{if } \rho_r \geq \rho_{\text{th}} \text{ and } \Lambda \geq \Lambda_{\text{lo}} \end{cases} \quad (3.16)$$

where $\rho_{\min}$ is the minimum resource threshold for ML execution, ρ_{th} is the RSU capacity threshold, and Λ_{lo} is the low-threat boundary. Both modes share the same attack signature definitions from Table 3.1; they differ in detection and mitigation depth.

...

3.4.2 Lightweight Mode: Rule-Based Signature Detection

In lightweight mode, each tier applies deterministic rules derived directly from Table 3.1. No ML inference or cryptographic key operations are performed. The

Algorithm 2 Per-Variant Detection Logic — Control-Plane Attacks (v_5 – v_6).

```

1: procedure DETECTCONTROLPLANE( $r, t, \theta_g, \mathcal{R}, \mathcal{M}$ )
2:   //  $v_5$ : Malicious RSU
3:   Compute  $\mathcal{S}_5(r, t)$  per Eq. (??) at SDN controller and vehicle monitors
4:    $\delta_{RSU} \leftarrow \frac{1}{|\mathcal{R}|} \sum_{r' \neq r} \mathcal{S}_5(r', t)$  ▷ cross-RSU disagreement
5:   if  $\mathcal{S}_5(r, t) > \theta_5^L$  and  $|\mathcal{S}_5(r, t) - \delta_{RSU}| > \epsilon_5$  then
6:     Isolate RSU  $r$ ; redistribute its clients to adjacent RSUs
7:     if  $\mathcal{M} = \mathcal{M}_F$  then
8:       Trigger Dilithium re-keying among remaining RSUs per Eq. (??)
9:     end if
10:    return  $v_5$ 
11:  end if
12:  //  $v_6$ : Malicious SDN Controller
13:  Compute  $\mathcal{S}_6(t)$  per Eq. (??) via RSU majority vote
14:   $D_{FL} \leftarrow \|\theta_g^{(SDN)}(t) - \frac{1}{|\mathcal{R}|} \sum_r \theta_r(t)\|_2$ 
15:  if  $\mathcal{S}_6(t) > \theta_6^L$  or  $D_{FL} > \theta_{FL}$  then
16:    RSU majority bypasses SDN controller for aggregation
17:    Activate MPC reconstruction per Eqs. (??)–(??)
18:    Alert network operator; IPFSREVOKE(SDN,  $t, \mathcal{M}_F, \mathcal{R}_{SDN}$ )
19:    return  $v_6$ 
20:  end if
21:  return legitimate
22: end procedure

```

detection decision at node v_i for attack variant a is:

$$D_{\text{LW}}^{(a)}(v_i, t) = \mathbf{1}[S^{(a)}(v_i, t) > \theta^{(a)}] \quad (3.17)$$

where $S^{(a)}(v_i, t)$ is the signature score for variant a computed from Table 3.1 and $\theta^{(a)}$ is the empirically set threshold for that variant. The composite lightweight detection flag is:

$$D_{\text{LW}}(v_i, t) = \bigvee_{a \in \mathcal{A}} D_{\text{LW}}^{(a)}(v_i, t) \quad (3.18)$$

where $\mathcal{A}$ is the set of six attack variants.

Lightweight Mitigation: Upon $D_{\text{LW}}(v_i, t) = 1$, the RSU immediately suspends beacon relay for v_i and broadcasts a local alert to neighboring RSUs. No cryptographic re-keying is performed in this mode. The alert message carries a time-stamped signature score vector $\mathbf{s} = [S^{(a)}]_{a \in \mathcal{A}}$ to allow the SDN controller to escalate to full mode if warranted.

3.4.3 Full Mode: ML-FL and LLM-Based Detection

In full mode, detection operates across all three tiers.

Tier 1 — Vehicle Nodes (ML-FL Clients): Each vehicle node maintains three lightweight FL client models: RSSI analyzer f_{RSSI} , temporal data analyzer f_{temp} , and trust score analyzer f_{trust} . The local detection score at vehicle v_i is:

$$\hat{y}_i = \sigma(\mathbf{w}_{\text{RSSI}}^\top \phi_{\text{RSSI}}(v_i) + \mathbf{w}_{\text{temp}}^\top \phi_{\text{temp}}(v_i) + \mathbf{w}_{\text{trust}}^\top \phi_{\text{trust}}(v_i)) \quad (3.19)$$

Algorithm 3 Lightweight Rule-Based Sybil Signature Detection (LW-SSD)

```

1: procedure LW-SSD( $v_i, t, \mathcal{O}, \theta^{(\cdot)}$ )
2:    $\mathbf{s} \leftarrow \mathbf{0}_{|\mathcal{A}|}$ 
3:    $S^{(\text{out})} \leftarrow \mathbf{1}[\tau_i \notin \mathcal{T}_{\text{valid}}]$ 
4:    $S^{(\text{sim})} \leftarrow \max_{j \neq i} \Phi_{\text{coloc}}(i, j)$  using (3.3)
5:    $S^{(\text{nsim})} \leftarrow \mathbf{1}[H_{\text{IAT}}(i)\theta_H]$  using (3.4)
6:    $S^{(\text{ind})} \leftarrow \max_j \text{sim}_{\text{DTW}}(i, j)$  using (3.5)
7:    $S^{(\text{rsu})} \leftarrow \mathbf{1}[\text{approve}(r, i) = 1 \wedge \nexists k : \text{RSSI}_{r,i} \in \mathcal{V}_k]$ 
8:    $S^{(\text{sdn})} \leftarrow \mathbf{1}[\Delta_{\text{div}} > \theta_{\text{div}}]$  using (3.6)
9:    $\mathbf{s} \leftarrow [S^{(\text{out})}, S^{(\text{sim})}, S^{(\text{nsim})}, S^{(\text{ind})}, S^{(\text{rsu})}, S^{(\text{sdn})}]$ 
10:   $D_{\text{LW}} \leftarrow \bigvee_{a \in \mathcal{A}} \mathbf{1}[s_a > \theta^{(a)}]$ 
11:  if  $D_{\text{LW}} = 1$  then
12:    Suspend beacon relay for  $v_i$ ; broadcast alert with  $\mathbf{s}$ 
13:  end if
14:  return  $D_{\text{LW}}, \mathbf{s}$ 
15: end procedure

```

where $\phi_{(\cdot)}$ are feature extractors and σ is the sigmoid activation. Vehicle nodes upload encrypted gradients, not raw data.

Tier 2 — RSU/Edge Servers (LLM Agents): The LLM agent at RSU r takes as input the concatenation of ML score vectors from associated vehicles and produces a contextual detection decision with reasoning:

$$(d_r, \text{reason}_r) = \text{LLM}_r(\mathbf{c}_r), \quad \mathbf{c}_r = [\hat{y}_i \parallel \phi_{\text{meta}}(v_i)]_{v_i \in \mathcal{V}_r} \quad (3.20)$$

where $\mathcal{V}_r$ is the set of vehicles associated with RSU r and ϕ_{meta} includes temporal, spatial and identity metadata.

Multi-Agent Coordination: Decisions from all RSU agents are aggregated at the SDN controller using a weighted consensus function:

$$D_{\text{global}}(v_i) = \mathbf{1} \left[\sum_{r \in \mathcal{R}} \omega_r \cdot d_r(v_i) \geq \theta_{\text{consensus}} \right] \quad (3.21)$$

where ω_r is the trust weight of RSU r (updated by the decentralized monitoring system) and $\theta_{\text{consensus}}$ is the consensus threshold.

3.4.4 Trust Score Formulation

The trust score $\mathcal{T}(v_i, t)$ for vehicle v_i is a multi-factor composite defined as:

$$\mathcal{T}(v_i, t) = \alpha \cdot \mathcal{T}_{\text{RSSI}}(v_i, t) + \beta \cdot \mathcal{T}_{\text{behav}}(v_i, t) + \gamma \cdot \mathcal{T}_{\text{hist}}(v_i, t) \quad (3.22)$$

where $\alpha + \beta + \gamma = 1$. The RSSI-based trust component is:

$$\mathcal{T}_{\text{RSSI}}(v_i, t) = 1 - \frac{1}{|\mathcal{O}|} \sum_{k \in \mathcal{O}} \Phi_{\text{coloc}}(i, \hat{s}_k, t) \quad (3.23)$$

where $\hat{s}_k$ is a known Sybil candidate and Φ_{coloc} is from (3.3). The behavioral trust component captures beacon rate anomaly:

$$\mathcal{T}_{\text{behav}}(v_i, t) = \exp \left(-\lambda \cdot \left| \frac{\bar{r}_i - r_{\text{nom}}}{r_{\text{nom}}} \right| \right) \quad (3.24)$$

Algorithm 4 Full-Mode ML+LLM Sybil Detection Pipeline (FM-SDP)

```

1: procedure FM-SDP( $v_i, \mathcal{V}_r, \mathcal{R}, \mathbf{w}_{\text{global}}, \theta_{\text{consensus}}$ )
2:   // Tier 1: Vehicle-side ML scoring
3:   for each  $v_i \in \mathcal{V}_r$  do
4:      $\hat{y}_i \leftarrow \sigma(\mathbf{w}_{\text{RSSI}}^\top \phi_{\text{RSSI}}(v_i) + \mathbf{w}_{\text{temp}}^\top \phi_{\text{temp}}(v_i) + \mathbf{w}_{\text{trust}}^\top \phi_{\text{trust}}(v_i))$    using (3.19)
5:     Upload encrypted gradient  $\tilde{\mathbf{g}}_i$  to RSU  $r$ 
6:   end for
7:   // Tier 2: RSU LLM agent inference
8:    $\mathbf{c}_r \leftarrow [\hat{y}_i \parallel \phi_{\text{meta}}(v_i)]$  for  $v_i \in \mathcal{V}_r$ 
9:    $(d_r, \text{reason}_r) \leftarrow \text{LLM}_r(\mathbf{c}_r)$  using (3.20)
10:  Forward  $(d_r, \omega_r)$  to SDN controller
11:  // Tier 3: SDN global consensus
12:   $D_{\text{global}}(v_i) \leftarrow \mathbf{1}[\sum_{r \in \mathcal{R}} \omega_r d_r(v_i) \geq \theta_{\text{consensus}}]$  using (3.21)
13:  Update  $\mathbf{w}_{\text{global}}$  via trimmed mean using (3.28)
14:  if  $D_{\text{global}}(v_i) = 1$  then
15:    Trigger PQC-REVOKE( $v_i$ ) (Algorithm 3)
16:  end if
17:  return  $D_{\text{global}}(v_i)$ 
18: end procedure

```

where $\bar{r}_i$ is the observed beacon rate, r_{nom} is the nominal rate, and λ is a decay constant. The historical trust uses an exponential moving average:

$$\mathcal{T}_{\text{hist}}(v_i, t) = \mu \cdot \mathcal{T}(v_i, t-1) + (1 - \mu) \cdot \mathcal{T}(v_i, t) \quad (3.25)$$

3.4.5 Federated Learning Objective and Aggregation

The global FL objective across all vehicle clients is:

$$\min_{\mathbf{w}} \mathcal{L}_{\text{global}}(\mathbf{w}) = \sum_{i=1}^N \frac{n_i}{n} \mathcal{L}_i(\mathbf{w}) \quad (3.26)$$

where n_i is the local dataset size of client i , $n = \sum_i n_i$, and $\mathcal{L}_i$ is the local cross-entropy loss. Each client uploads an encrypted gradient $\tilde{\mathbf{g}}_i = \mathbf{g}_i + \mathbf{n}_i$ where $\mathbf{n}_i \sim \mathcal{N}(0, \sigma_{\text{dp}}^2 \mathbf{I})$ is differential privacy noise. The mid-tier RSU aggregation is:

$$\mathbf{w}_r^{(t+1)} = \mathbf{w}_r^{(t)} - \eta \sum_{i \in \mathcal{V}_r} \frac{n_i}{n_r} \tilde{\mathbf{g}}_i^{(t)} \quad (3.27)$$

The global SDN aggregation applies Byzantine-robust trimmed mean:

$$\mathbf{w}_{\text{global}}^{(t+1)} = \frac{1}{|\mathcal{R}| - 2k} \sum_{r \in \mathcal{R} \setminus \mathcal{R}_k^\pm} \mathbf{w}_r^{(t)} \quad (3.28)$$

where $\mathcal{R}_k^\pm$ removes the k highest and lowest norm updates, providing robustness against a malicious SDN controller attempting to poison the global model.

3.4.6 Full Mitigation: PQC-Based Identity Revocation

Upon $D_{\text{global}}(v_i) = 1$, full mitigation is triggered. We propose a **threshold ring signature** scheme built on CRYSTALS-Dilithium [?] for distributed revocation authorization, combined with CRYSTALS-Kyber for session re-keying.

Threshold Ring Signature Scheme: Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the set of RSUs acting as ring members. A revocation decision requires a threshold t -of- n RSU co-signatures. Each RSU r_j holds a Dilithium key pair $(\text{pk}_j, \text{sk}_j) \leftarrow \text{Dilithium.KeyGen}(\lambda)$. A partial revocation signature on node identity ID_i is:

$$\sigma_j = \text{Dilithium.Sign}(\text{sk}_j, m_{\text{rev}}) \quad (3.29)$$

where $m_{\text{rev}} = H(\text{ID}_i \parallel t_{\text{rev}} \parallel \Delta_{\text{evidence}})$ and H is SHA3-256. The aggregated threshold signature is valid if:

$$\text{Dilithium.Verify}\left(\bigoplus_{j \in \mathcal{S}} \text{pk}_j, m_{\text{rev}}, \bigoplus_{j \in \mathcal{S}} \sigma_j\right) = 1, \quad |\mathcal{S}| \geq t \quad (3.30)$$

where $\mathcal{S}$ is the subset of consenting RSUs and $\bigoplus$ denotes the lattice-compatible aggregation over the Dilithium module lattice $\mathcal{M}_q^{k \times l}$.

CRYSTALS-Dilithium is selected for threshold revocation signatures because it is an NIST PQC standard (FIPS 204) offering compact signatures (≈ 2420 bytes at security level 2) suitable for RSU-to-SDN communication. CRYSTALS-Kyber (FIPS 203) provides IND-CCA2-secure key encapsulation for session re-keying with ciphertexts of ≈ 768 bytes at level 2, compatible with vehicular OBU memory constraints. MPC via additive secret sharing requires only $O(n)$ communication rounds, matching the RSU-as-edge-server architecture where RSUs naturally serve as the MPC parties. The combination is post-quantum secure under the Module-LWE and Module-SIS hardness assumptions.

Session Re-keying via Kyber KEM: After revocation, the SDN controller initiates Kyber-based re-keying for all legitimate vehicles in the affected RSU zone. The key encapsulation produces:

$$(K, c) \leftarrow \text{Kyber.Encaps}(\text{pk}_{\text{vehicle}}) \quad (3.31)$$

where $K \in \{0, 1\}^{256}$ is the shared session key and $c \in \mathcal{R}_q^k$ is the ciphertext. The vehicle decapsulates:

$$K' = \text{Kyber.Decaps}(\text{sk}_{\text{vehicle}}, c) \quad (3.32)$$

A successful re-key is confirmed when $K' = K$, after which the revoked identity ID_i is excluded from all subsequent sessions.

MPC-Based Secure Evidence Aggregation: To avoid exposing raw detection evidence to any single RSU, the evidence vectors $\mathbf{e}_r$ from each RSU are aggregated using additive secret sharing. Each RSU r_j splits its evidence into n shares:

$$\mathbf{e}_j = \sum_{k=1}^n s_{jk}, \quad s_{jk} \leftarrow \mathcal{U} \quad (3.33)$$

The reconstructed aggregate evidence $\hat{\mathbf{E}}$ at the SDN controller is:

$$\hat{\mathbf{E}} = \sum_{j=1}^n \mathbf{e}_j = \sum_{j=1}^n \sum_{k=1}^n s_{jk} \quad (3.34)$$

Algorithm 5 PQC Threshold Revocation (PQC-REVOKE)

```
1: procedure PQC-REVOKE( $ID_i, \mathcal{R}, t$ )
2:    $m_{\text{rev}} \leftarrow H(ID_i \parallel t_{\text{rev}} \parallel \hat{\mathbf{E}})$  using MPC aggregate (3.34)
3:    $\mathcal{S} \leftarrow \emptyset$ 
4:   for each RSU  $r_j \in \mathcal{R}$  do
5:      $\sigma_j \leftarrow \text{Dilithium.Sign}(\text{sk}_j, m_{\text{rev}})$  using (3.29)
6:     Broadcast  $\sigma_j$  to SDN controller
7:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{j\}$ 
8:     if  $|\mathcal{S}| \geq t$  then break
9:   end if
10:  end for
11:  Verify:  $\text{Dilithium.Verify}(\bigoplus_{j \in \mathcal{S}} \text{pk}_j, m_{\text{rev}}, \bigoplus_{j \in \mathcal{S}} \sigma_j)$  using (3.30)
12:  if valid then
13:    Publish  $\text{CID}_j$  for each  $j \in \mathcal{S}$  to IPFS using (3.35)
14:    SDN controller pins manifest  $\mathcal{M}_{\text{rev}}(ID_i)$  using (3.36)
15:    Initiate Kyber re-keying:  $(K, c) \leftarrow \text{Kyber.Encaps}(\text{pk}_{\text{vehicle}})$  for all  $v \in \mathcal{V} \setminus \{ID_i\}$  using (3.31)
16:    Broadcast revocation alert with  $\mathcal{M}_{\text{rev}}$ 
17:  else
18:    Log failure; escalate to manual review
19:  end if
20: end procedure
```

No individual RSU learns the evidence contributed by others, preserving inter-RSU privacy during joint revocation decisions.

...

3.4.7 IPFS-Based Real-Time Evidence Logging

Upon generating a revocation decision, each participating RSU r_j publishes its local evidence vector $\mathbf{e}_j$ to IPFS, producing a content identifier:

$$\text{CID}_j = H_{\text{IPFS}}(\mathbf{e}_j \parallel \sigma_j \parallel t_{\text{rev}}) \quad (3.35)$$

where H_{IPFS} is the multihash function (SHA2-256 by default in IPFS). The SDN controller maintains a pinned manifest:

$$\mathcal{M}_{\text{rev}}(ID_i) = \{\text{CID}_j\}_{j \in \mathcal{S}}, \quad |\mathcal{S}| \geq t \quad (3.36)$$

The manifest is broadcast to all RSUs, which verify CID authenticity before accepting the revocation. This provides tamper-evident, decentralized evidence accumulation without blockchain consensus latency. Real-time streaming of evidence is achieved via IPFS pubsub, where RSUs subscribe to the topic “sybil-evidence- ID_i ” and receive updates within the RSU communication range.

...

3.4.8 Dual-Mode Temporal Operation

Table 3.3 classifies each component by its operational latency class. The end-to-end detection latency in full mode is modeled as:

$$L_{\text{full}} = L_{\text{infer}} + L_{\text{LLM}} + L_{\text{consensus}} + L_{\text{revoke}} \quad (3.37)$$

Algorithm 6 IPFS Evidence Publication and Revocation Propagation (IPFS-EVP)

```

1: procedure IPFS-EVP( $ID_i, \mathcal{S}, \mathbf{e}, \sigma$ )
2:   for each RSU  $r_j \in \mathcal{S}$  do
3:      $CID_j \leftarrow H_{IPFS}(\mathbf{e}_j \parallel \sigma_j \parallel t_{rev})$  using (3.35)
4:     Pin  $CID_j$  to local IPFS node
5:     Publish  $CID_j$  on pubsub topic “sybil-evidence- $ID_i$ ”
6:   end for
7:    $\mathcal{M}_{rev} \leftarrow \{CID_j\}_{j \in \mathcal{S}}$ 
8:   SDN controller broadcasts  $\mathcal{M}_{rev}$  to all RSUs
9:   for each RSU  $r_k \in \mathcal{R}$  do
10:    Retrieve and verify each  $CID_j \in \mathcal{M}_{rev}$ 
11:    Update local blacklist:  $\mathcal{BL} \leftarrow \mathcal{BL} \cup \{ID_i\}$ 
12:   end for
13:   return  $\mathcal{M}_{rev}$ 
14: end procedure

```

Table 3.3: Temporal Classification of Framework Components

Component	Mode	Operation
Rule-based signature check	Both	Real-time (< 10 ms)
RSSI / trust scoring	Full	Real-time (< 50 ms)
ML-FL local inference	Full	Real-time (< 100 ms)
LLM agent inference	Full	Near-real-time (< 500 ms)
IPFS evidence streaming	Full	Real-time (pubsub)
Dilithium threshold signing	Full	Near-real-time (< 200 ms)
Kyber session re-keying	Full	Near-real-time (< 100 ms)
FL global aggregation	Both	Offline (per round)
LLM LoRA fine-tuning	Both	Offline (periodic)

where L_{infer} is the vehicle-tier ML inference time, L_{LLM} is the RSU LLM agent response time, $L_{consensus}$ is the multi-agent SDN consensus delay, and L_{revoke} is the Dilithium threshold signing and IPFS publication time. The lightweight mode latency reduces to:

$$L_{LW} = L_{rule} \ll L_{full}, \quad L_{rule} < 10 \text{ ms} \quad (3.38)$$

...

3.5 Evaluation Metrics

3.5.1 External Baselines

Three external baselines are selected addressing the same general Sybil attack class in vehicular networks:

B1 — FLEMDS [?]: FL-based Sybil detection in SDVF using fuzzy logic client selection; represents FL-only detection without LLM reasoning or PQC mitigation (87% detection accuracy).

B2 — Benadla et al. [?]: RSSI-based distance estimation with distributed ledger evidence in vehicular fog; represents hybrid physical-layer and cryptographic Sybil detection without federated AI.

B3 — Baza et al. [?]: Proof-of-work and proof-of-location Sybil detection in VANETs (IEEE Transactions on Dependable and Secure Computing); represents position-proof cryptographic detection without AI or federation.

3.5.2 Internal Ablation Baselines

A1: Full system without LLM agents (ML-FL only).

A2: Full system without PQC mitigation (no Dilithium or Kyber).

A3: Lightweight mode only (no full-mode activation).

A4: Full mode with standard FedAvg instead of MPC aggregation.

3.5.3 Evaluation Metrics

Metrics are organized across four evaluation dimensions.

Network-Level Routing Metrics:

M1 — Packet Delivery Ratio (PDR): Ratio of successfully delivered packets to total transmitted packets, measured under attack and post-mitigation; quantifies Sybil-induced routing disruption and its restoration after revocation.

M2 — End-to-End Routing Latency (ms): Average packet delivery delay under active Sybil attack versus post-mitigation baseline.

Sybil Attack Impact Metrics:

M3 — Packet Attraction Ratio: Proportion of network traffic diverted through Sybil nodes; measures the severity of Sybil-induced routing manipulation.

M4 — Congestion Level: Network congestion induced by Sybil node injection of false traffic information.

Security Detection Metrics:

M5 — Matthews Correlation Coefficient (MCC): Primary detection quality metric robust to class imbalance across six attack variants; captures true and false positives and negatives in a single scalar, replacing per-variant F1-score.

M6 — False Positive Rate (FPR): Proportion of legitimate nodes incorrectly flagged as Sybil; directly impacts road safety reliability.

M7 — Revocation Latency (ms): End-to-end time from detection decision to completed threshold Dilithium revocation and Kyber re-keying under both $\mathcal{M}_L$ and $\mathcal{M}_F$.

System Efficiency Metrics:

M8 — Communication Overhead (KB per detection event): Bandwidth cost of IPFS evidence publication and threshold signature coordination across RSU tiers.

M9 — Model Convergence Rounds: Number of FL aggregation rounds required for the global model to converge, evaluating the efficiency of MPC-secured federated aggregation.

M10 — Computational Complexity per Detection Event: FLOPs and wall-clock inference time per node per detection cycle, measured separately for $\mathcal{M}_L$ and $\mathcal{M}_F$ at each tier; validates dual-mode feasibility under OBU and RSU resource constraints per Eqs. (3.39)–(3.41).

...

3.5.4 Complexity Analysis

Let N_v be the number of vehicle nodes per RSU, d the feature dimension, W the RSSI observation window, L the LLM context length, and p the Dilithium polynomial dimension.

Lightweight mode ($\mathcal{M}_L$): Signature score computation per node requires $\mathcal{O}(W)$ for RSSI cross-correlation (Eq. (??)), $\mathcal{O}(W \log W)$ for DTW (Eq. (??)), and $\mathcal{O}(1)$ for threshold comparison (Eq. (??)). Total per-cycle complexity at the vehicle tier:

$$\mathcal{C}_L = \mathcal{O}(N_v \cdot W \log W) \quad (3.39)$$

Full mode ($\mathcal{M}_F$): ML model inference is $\mathcal{O}(d^2)$ per node; LLM multi-head attention over context L is $\mathcal{O}(L^2 \cdot d_{model})$; threshold Dilithium signing is $\mathcal{O}(p \log p)$ per RSU per NTT operation; Kyber encapsulation is $\mathcal{O}(p \log p)$. Total per-cycle complexity at the RSU tier:

$$\mathcal{C}_F = \mathcal{O}(N_v d^2 + |\mathcal{A}| L^2 d_{model} + t \cdot p \log p) \quad (3.40)$$

Complexity ratio:

$$\mathcal{CR} = \frac{\mathcal{C}_F}{\mathcal{C}_L} = \mathcal{O}\left(\frac{N_v d^2 + |\mathcal{A}| L^2 d_{model} + t p \log p}{N_v W \log W}\right) \quad (3.41)$$

$\mathcal{CR}$ quantifies the computational cost of activating full mode relative to lightweight mode and is evaluated empirically across vehicle densities and mobility scenarios to validate mode-selection thresholds θ_{res} and T_{min} in Eq. (??).

Chapter 4

Timeline and Resource Required

4.1 Timeline

Contributors are represented as below.

- A - S.A.D.N. Maduwantha
- B - N.A.K.D. Nissanka
- C - E.W.G.I.S. Ranasinghe
- D - O.K. Ranathunga

Project Details		Project Weeks																															
Task	Contributor	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Project Selection	A,B,C,D																																
Literature Review and Background Study	A,B,C,D																																
Proposal Creation and Submission	A,B,C,D																																
SDVEN simulation and Sybil attack Modelling (NS3)	A,B																																
Multi-agent System Architecture Design	C,D																																
Federated learning framework (LLM-based)	A,B																																
LLM fine-tuning for Sybil patterns	C,D																																
Real-time detection and agent Coordination	A,B																																
Testing, optimization and evaluation	A,B,C,D																																
Cross-domain transfer to GEM fraud detection	C,D																																
Final report, documentation and viva	A,B,C,D																																

Figure 4.1: Project Timeline and Task Allocation

4.2 Resource Required

1. Programming Languages

Python will be used as the primary programming language for ML, Agent design and FL developments. C++ will be used to program NS-3 simulations.

2. Simulation Tools

Network Simulator 3 (NS-3) will be used to simulate Software Defined Vehicular Edge Networks and Sybil attack scenarios

3. LLM Tools

Hugging Face Transformers, Mistral and Llama light weight models will be used to develop and integrate Large Language models for multi agent implementations.

4. FL Frameworks

Across the distributed nodes, hope to use PyTorch, TensorFlow and Flower Python frameworks to implement FL flows.

5. Supporting Libraries

NumPy, Pandas, Scikit-learn and model evaluation libraries are the supporting libraries used in processing data, extracting features, and evaluating performance.

6. Hardware Requirements

GPU-enabled computing resources used for Simulations, LLM model fine tuning, ML model training and checking the FL model aggregations.

4.3 Budget Estimation

The proposed budget is minimal and cost-effective, focusing mainly on cloud-based computational resources and AI model access. Paid services such as Google Colab Pro and LLM APIs are essential for efficient experimentation, scalability, and validation of the proposed federated multi-agent Sybil attack detection framework.

1. High-Performance Cloud Compute (Google Colab Pro Subscription)

- Usage: Training of federated learning agents, large-scale graph neural network processing for Sybil detection, and long-running model validation.
- Expected Duration: Equivalent monthly credits for 5 months.
- **Total Estimated Cost (LKR): 15,000 LKR**

2. OpenAI API Usage (LLM Integration)

- Usage: Fine-tuning detection prompts and multi-agent communication simulation.
- API Calls: Inference and embedding generation for behavior analysis.
- **Total Estimated Cost (LKR): 15,000 LKR**

Table 4.1: Estimated Budget of the Project

Element No	Element Name	Estimated Cost (LKR)
1	Google Colab Pro Subscription	15,000
2	OpenAI API Usage	15,000
Total	Overall Budget	30,000 LKR

Chapter 5

Conclusion

This proposal presents an optimized federated large language model-based multi-agent framework for detecting Sybil attacks in software-defined vehicular edge networks. The proposed approach addresses critical security challenges in SD-VENs by using decentralized, privacy-preserving and intelligent attack detection across both data and control planes. Using federated learning with large language models and cooperative multi-agent intelligence allows vehicles, RSUs, and the SDN controller to collaboratively assess trust or behavior and then detect malicious nodes without relying on a single centralized controller.

The proposed system has decentralized trust and behavior monitoring and considers a different types of Sybil attacks types such as outsider, insider direct simultaneous, insider direct non-simultaneous, insider indirect and also malicious infrastructure entities such as RSUs and SDN controllers. Behavioral analysis, data similarity assessment, temporal analysis, RSSI-based detection and authentication mechanisms are used to enhance detection accuracy under high mobility vehicular conditions. Furthermore, the cross-domain transfer learning enables the learned representations from vehicular Sybil attack detection to be effectively use to gem fraud detection which represents the generalizability of the proposed framework.

Overall, this research is expected to contribute a scalable, accurate and resilient security solution for software defined vehicular edge networks while extending its applicability to other fraud detection domains.

References

- [1] N. H. Hussein, S. P. Koh, C. T. Yaw, S. K. Tiong, F. Benedict, T. Yusaf, K. Kadirgama, and T. C. Hong, “Sdn-based vanet routing: A comprehensive survey on architectures, protocols, analysis, and future challenges,” *IEEE Access*, 2024.
- [2] S. A. Ul Haq, M. Imran, N. Shah, and G.-M. Muntean, “Sdn-based edge computing in vehicular communication networks: A survey of existing approaches,” *IEEE Access*, 2025.
- [3] K. Wang, H. Yin, W. Quan, and G. Min, “Enabling collaborative edge computing for software defined vehicular networks,” *IEEE network*, vol. 32, no. 5, pp. 112–117, 2018.
- [4] R. R and S. S, “A survey on security challenges and threats of vehicular adhoc networks(vanets),” *International Journal of Engineering Research & Technology*, vol. 3, 2 2014.
- [5] W. B. Jaballah, M. Conti, and C. Lal, “A survey on software-defined vanets: benefits, challenges, and future directions,” *arXiv preprint arXiv:1904.04577*, 2019.
- [6] L. Nkenyereye, L. Nkenyereye, S. R. Islam, C. A. Kerrache, M. Abdullah-Al-Wadud, and A. Alamri, “Software defined network-based multi-access edge framework for vehicular networks,” *IEEE Access*, vol. 8, pp. 4220–4234, 2019.
- [7] L. Liu, C. Chen, Q. Pei, S. Maharjan, and Y. Zhang, “Vehicular edge computing and networking: A survey,” *Mobile networks and applications*, vol. 26, no. 3, pp. 1145–1168, 2021.
- [8] T. Zaidi and S. Faisal, “An overview: Various attacks in vanet,” in *2018 4th International Conference on Computing Communication and Automation (ICCCA)*, pp. 1–6, IEEE, 2018.
- [9] J. Newsome, E. Shi, D. Song, and A. Perrig, “The sybil attack in sensor networks: analysis & defenses,” in *Proceedings of the 3rd international symposium on Information processing in sensor networks*, pp. 259–268, 2004.
- [10] S. Benadla and O. R. Merad-Boudia, “The impact of sybil attacks on vehicular fog networks,” in *2021 International Conference on Recent Advances in Mathematics and Informatics (ICRAMI)*, pp. 1–6, IEEE, 2021.

- [11] R. P. Nayak, S. K. Bhoi, K. S. Sahoo, S. Sethi, S. Mohapatra, and M. Bhuyan, "Unveiling sybil attacks using ai-driven techniques in software-defined vehicular networks," *Security and Privacy*, vol. 8, no. 1, p. e487, 2025.
- [12] J. R. Douceur, "The sybil attack," in *International workshop on peer-to-peer systems*, pp. 251–260, Springer, 2002.
- [13] S. Capkun and J.-P. Hubaux, "Secure positioning of wireless devices with application to sensor networks," in *Proceedings IEEE 24th Annual Joint Conference of the IEEE Computer and Communications Societies.*, vol. 3, pp. 1917–1928, IEEE, 2005.
- [14] A. Tondwalkar and P. Vinayakray-Jani, "Secure localisation of wireless devices with application to sensor networks using steganography," *Procedia Computer Science*, vol. 78, pp. 610–616, 2016.
- [15] M. Raya and J.-P. Hubaux, "Securing vehicular ad hoc networks," *Journal of computer security*, vol. 15, no. 1, pp. 39–68, 2007.
- [16] X. Lin, X. Sun, P.-H. Ho, and X. Shen, "Gsis: A secure and privacy-preserving protocol for vehicular communications," *IEEE Transactions on vehicular technology*, vol. 56, no. 6, pp. 3442–3456, 2007.
- [17] C. Zhang, R. Lu, X. Lin, P.-H. Ho, and X. Shen, "An efficient identity-based batch verification scheme for vehicular sensor networks," in *IEEE INFOCOM 2008-The 27th Conference on Computer Communications*, pp. 246–250, IEEE, 2008.
- [18] D. E. Laouiti, M. Ayaida, N. Messai, S. Najeh, L. Najjar, and F. Chaabane, "Sybil attack detection in vanets using an adaboost classifier," in *2022 International Wireless Communications and Mobile Computing (IWCMC)*, pp. 217–222, IEEE, 2022.
- [19] S. Kakulla and S. Malladi, "Sybil attack detection in vanet using machine learning approach," *Ingenierie des Systemes d'Information*, vol. 27, no. 4, p. 605, 2022.
- [20] S. Azam, M. Bibi, R. Riaz, S. S. Rizvi, and S. J. Kwon, "Collaborative learning based sybil attack detection in vehicular ad-hoc networks (vanets)," *Sensors*, vol. 22, no. 18, p. 6934, 2022.
- [21] N. Mageshkumar and J. Bapu, "Sybil attack detection in vanet using cnn based lstm framework model,"
- [22] K. ., S. A, B. Kasi, and A. A, "Sybil attack detection in vanet using cnn enhanced with chaotic maps and elephant herding optimization for secure data transmission," *Journal of Machine and Computing*, pp. 1233–1247, 04 2025.
- [23] T. P. Nguyen, H. Nam, and D. Kim, "Transformer-based attention network for in-vehicle intrusion detection," *IEEE Access*, vol. 11, pp. 55389–55403, 2023.

- [24] W. Khan, J. Ahmad, N. Alasbali, A. Al Mazroa, M. S. Alshehri, and M. S. Khan, "A novel transformer-based explainable ai approach using shap for intrusion detection in vehicular ad hoc networks," *Computer Networks*, p. 111575, 2025.
- [25] A. M. Elbir, B. Soner, S. Çöleri, D. Gündüz, and M. Bennis, "Federated learning in vehicular networks," in *2022 IEEE International Mediterranean Conference on Communications and Networking (MeditCom)*, pp. 72–77, IEEE, 2022.
- [26] Z. Du, C. Wu, T. Yoshinaga, K.-L. A. Yau, Y. Ji, and J. Li, "Federated learning for vehicular internet of things: Recent advances and open issues," *IEEE Open Journal of the Computer Society*, vol. 1, pp. 45–61, 2020.
- [27] A. Muhammad, "Real-time threat detection in vehicular networks using federated learning," 2025.
- [28] D. B. Naik and V. Dondeti, "Trust-based secure federated learning framework to mitigate internal attacks for intelligent vehicular networks," *Peer-to-Peer Networking and Applications*, vol. 18, no. 2, p. 10, 2025.
- [29] M. Asad and S. Otoum, "Fl-sats: Federated learning for sybil attack detection in transportation system," in *ICC 2025-IEEE International Conference on Communications*, pp. 3376–3381, IEEE, 2025.
- [30] L. J. Vinita and V. Vetriselvi, "Federated learning-based misbehaviour detection on an emergency message dissemination scenario for the 6g-enabled internet of vehicles," *Ad Hoc Networks*, vol. 144, p. 103153, 2023.